INDEFINITE INTEGRATION

SYNOPSIS

Defination :- The inverse process of differentiation is called integration.

Let $F(x)$ be a differential function of x such that primitive (or) antiderivative (or) an indefitine integral (or) simply integral of $f(x)$ w. r. to 'x' and is written symbolically as

$$\int f(x)dx = F(x)+c \Leftrightarrow \frac{d}{dx}\left(F(x)+c\right) = f(x)$$

Where $f(x)$ is called the integrand and 'c' is called the constant of integration (Where 'c' may be real (or) imaginary)

General Formulae :

- $\frac{d}{dx}(F(x)) = f(x) \Rightarrow \int f(x)dx = F(x)+c$
- If $K \in R$, $\int K dx = Kx + C$
- If $n \neq -1$, $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + c$
- $\int \frac{1}{x} dx = \log|x| + C$
- $\int \sqrt{x} dx = \frac{2}{3} x^{\frac{3}{2}} + C$
- $\int e^x .dx = e^x + C$
- $\int a^x \, dx = \frac{a^x}{\log a} + C$ (for a > 0, $a \neq 1$)
- $\int Sin\, x\, dx = -Cos\, x + C$
- $\int Cos\, x\, dx = Sin\, x + C$
- $\int \tan x\, dx = \log|Sec\, x| + C = -\log|\cos x| + c$
- $\int Cot\ x\ dx = \log|Sin\ x| + C = -\log|\operatorname{cosec} x| + c$
- $\int Sec\, x\, dx = \log|Sec\, x + \tan x| + C$

  $= \log\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + C$
- $\int Co\sec x\, dx = \log|Co\sec x - Cot\, x| + C$

  $= \log\left|\tan\frac{x}{2}\right| + C = -\log|Co\sec x + Cot\, x| + C$
- $\int Sec\, x \tan x\, dx = Sec x + C$
- $\int Co\sec x \cot x\, dx = -Co\sec x + C$
- $\int \sec^2 x\ dx = \tan x + C$
- $\int Co\sec^2 x\ dx = -Cot\, x + C$
- $\int Sin\, h\, x\, dx = Cos\, h\, x + C$
- $\int Cos\, h\, x\ dx = Sin\, h\, x + C$
- $\int \tan h\, x\ dx = \log|Cosh\, x| + C$
- $\int Cot\, h\, x\ dx = \log|Sin\, h\, x| + C$
- $\int Sec\, h\, x\ dx = 2\tan^{-1}\left(e^x\right) + C$
- $\int Co\sec hx = \log|\tan h\frac{x}{2}| + C$
- $\int Sec\, h^2 x\, dx = \tan h\, x + C$
- $\int Co\sec\ h^2 x\ \ dx = -Cot\ h\, x + C$
- $\int Sec\, hx \tan h\, x\, dx = -Sec\, hx + C$
- $\int Co\sec h\, x\, Cot\, hx\, dx = -Co\sec hx + C$
- $\int \frac{dx}{\sqrt{1-x^2}} = Sin^{-1}x + C = -COS^{-1}x + c$
- $\int \frac{dx}{1+x^2} = \tan^{-1} x + C = -Cot^{-1}x + c$
- $\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1}x + c,\ |x| > 1$

- $\int \frac{1}{|x|\sqrt{x^2-1}}dx = -\operatorname{cosec}^{-1}x + c,\ |x|>1$
- $\int \frac{1}{\sqrt{1+x^2}}dx = \sinh^{-1}x + c$

$= \log\left(x+\sqrt{x^2+1}\right)+c \quad \forall x \in R$

- $\int \frac{1}{\sqrt{x^2-1}}dx = \cosh^{-1}x + c\ \forall x \in (1,\infty)$

$= -\cosh^{-1}(-x)+c\ \forall x \in (-\infty,-1)$ (or)

$= \log\left|x+\sqrt{x^2-1}\right|+c\ \forall x \in (-\infty,-1)\cup(1,\infty)$

- $\int |x|\,dx = \frac{x|x|}{2}+c$
- $\int \frac{|x|}{x}dx = |x|+c, x\neq 0$

Properties :

- $f(x), g(x)$ are two integrable functions then $\int (f(x)\pm g(x))dx = \int f(x)dx \pm \int g(x)dx$
- If $f_1(x), f_2(x), ...f_n(x)$ integrable functions then $\int [f_1(x)+f_2(x)+.....+f_n(x)]dx =$ $\int f_1(x)dx + \int f_2(x)dx + ...+\int f_n(x)dx$
- If $f(x)$ is an integrable function and k is a number then $\int (kf(x))dx = k\int f(x)dx$
- If $f(x), g(x)$ are two integrable functions and k_1, k_2 are two real number then

$\int [k_1 f(x) \pm k_2 g(x)]dx = k_1\int f(x)dx \pm k_2 \int g(x)dx$

Important Results :

- $\int \left[\frac{d}{dx}f(x)\right]dx = f(x)+C$
- $\frac{d}{dx}\left(\int f(x)dx\right) = f(x)$
- $\int \{f(x)\}^n . f^1(x)dx = \frac{\{f(x)\}^{n+1}}{n+1}+C\ (n\neq -1)$
- $\int \frac{f^1(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)}+C$
- $\int \frac{f^1(x)}{f(x)}dx = \log|f(x)|+C$
- $\int f^1\{g(x)\}.g^1(x)dx = f\{g(x)\}+C$
- $\int f^1(ax+b)dx = \frac{f(ax+b)}{a}+C$
- $\int \frac{f^1(x)}{\sqrt{a^2-(f(x))^2}}dx = \text{Sin}^{-1}\left(\frac{f(x)}{a}\right)+\text{C}$
- $\int \frac{f^1(x)}{\sqrt{[f(x)]^2-a^2}}dx = \text{Cosh}^{-1}\left[\frac{f(x)}{a}\right]+\text{C}$

$= \log\left|f(x)+\sqrt{[f(x)]^2-a^2}\right|+C$

- $\int \frac{f'(x)}{\sqrt{[f(x)]^2+a^2}}dx = Sinh^{-1}\left[\frac{f(x)}{a}\right]+c$

$= \log\left|f(x)+\sqrt{[f(x)]^2+a^2}\right|+c$

- $\int \frac{f'(x)}{a^2+(f(x))^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{f(x)}{a}\right)+c$

Standard Integrals :

- $\int \frac{1}{\sqrt{a^2-x^2}}dx = \text{Sin}^{-1}\left(\frac{x}{a}\right)+c$

$\int \frac{1}{\sqrt{x^2-a^2}}dx = \cosh^{-1}\left(\frac{x}{a}\right)+c\ \forall x \in (a,\infty)$

$= -\cosh^{-1}\left(\frac{-x}{a}\right)+c\ \forall x \in (-\infty,-a)$ (or)

$= \log\left|x+\sqrt{x^2-a^2}\right|+c\ \forall x \in (-\infty,-a)\cup(a,\infty)$

- $\int \frac{1}{\sqrt{x^2+a^2}}dx = Sinh^{-1}\left(\frac{x}{a}\right)+c$

$= \log\left|x+\sqrt{x^2+a^2}\right|+c$

➢ $\int \frac{1}{x^2+a^2}dx = \frac{1}{a}\,\mathrm{Tan}^{-1}\left(\frac{x}{a}\right)+c$

➢ $\int \frac{1}{a^2-x^2}dx = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+c$

➢ $\int \frac{1}{x^2-a^2}dx = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+c$

➢ $\int \frac{1}{x\sqrt{x^2-a^2}}dx = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right)+c$

➢ $\int\sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}Sin^{-1}\left(\frac{x}{a}\right)+c$

➢ $\int\sqrt{x^2-a^2}\,dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}Cosh^{-1}\left(\frac{x}{a}\right)+c$

$= \frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+c$

➢ $\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\sinh^{-1}\left(\frac{x}{a}\right)+c$

$=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+c$

Integration By Parts :

➢ If u and v are functions of x, then

$\int uv\,dx = u\int v\,dx - \int\left(\frac{du}{dx}\right)\left(\int v\,dx\right)dx$ (or)

$\int f(x)\cdot g(x)dx = f(x)\int g(x)dx - \int\left[f'(x)\int g(x)dx\right]\cdot dx$

where f(x) = First function, g(x) = second function.

➢ Proper choice of first and second function :

a) The frist function is the function which comes first in the word ILATE.

b) If one of the two functions is not directly integrable, then take this function as the first function.

c) If one of the function is not directly integrable, and there is no other function, then unity is taken as the second function.

➢ If u and v are easily derivable, integrable functions of x, then $\int uv\,dx=uv_1-u^{I}v_2+u^{II}v_3-u^{III}v_4+...$

➢ $\int e^x f(x)dx=e^x\left(f(x)-f^{I}(x)+f^{II}(x)-f^{III}(x)+....\right)$

where f is a polynomial in x

➢ $\int x^n\cdot\log x\,dx=\frac{x^{n+1}}{n+1}\left[\log x-\frac{1}{n+1}\right]+c$ (for $n\neq -1$)

➢ $\int e^x\left[f(x)+f'(x)\right]dx = e^x f(x)+c$

➢ $\int e^{kx}\left[k\,f(x)+f^{1}(x)\right]dx = \int D\left(e^{kx}f(x)\right) = e^{kx}f(x)+c$

➢ $\int e^{-x}\left[f(x)-f'(x)\right]dx = -e^{-x}f(x)+C$

➢ $\int[f(x)+xf'(x)]dx = \int D(x\,f(x)) = xf(x)+c$

➢ $\int e^{ax}Sin(bx+c)dx=$

$\frac{e^{ax}}{a^2+b^2}[aSin(bx+c)-bCos(bx+c)]+K$

➢ $\int e^{ax}Cos(bx+c)dx=$

$\frac{e^{ax}}{a^2+b^2}$[aCos(bx+c)+bsin(bx+c)]+K

➢ $\int a^x\cdot Sin(bx+c)dx = \int e^{x\log a}.\sin(bx+c)\,dx$

$=\frac{a^x}{(\log a)^2+b^2}[(\log a)\cdot Sin(bx+c)-bCos(bx+c)]+K$

➢ $\int a^x Cos(bx+c)dx =$

$\frac{a^x}{(\log a)^2+b^2}[(\log a)\cdot\cos(bx+c)+bSin(bx+c)]+K$

➢ $\int xe^{ax}.\sin(bx+c)dx=\frac{xe^{ax}}{a^2+b^2}[a\sin(bx+c)-b\cos(bx+c)]$

$-\frac{e^{ax}}{(a^2+b^2)^2}\left[(a^2-b^2)\sin bx-2ab\cos bx\right]+k$

➢ $\int x.e^{ax}.\cos(bx+c)dx=\frac{x.e^{ax}}{a^2+b^2}[a\cos(bx+c)+b\sin(bx+c)]$

$-\frac{e^{ax}}{(a^2+b^2)^2}\left[(a^2-b^2)\cos(bx+c)+2ab\sin(bx+c)\right]+k$

Some Standard Substitutions :

➢ **Form of integrand :** $\sqrt{a^2-x^2}$, $\frac{1}{\sqrt{a^2-x^2}}$

Substitution : $x=a\sin\theta$ or $x=a\cos\theta$

➢ **Form of integrand :** $\sqrt{a^2+x^2}$, $\frac{1}{\sqrt{a^2+x^2}}$

Substitution : $x=a\tan\theta$ or $x=a\cot\theta$ or $x=a\sinh\theta$

- **Form of integrand :** $\frac{1}{\sqrt{x^2-a^2}}, \sqrt{x^2-a^2}$

  Substitution : $x=a\sec\theta$ (or) $x=a\,\mathrm{cosec}\,\theta$
  (or) $x=a\cosh\theta$

- **Form of integrand :** $\sqrt{\frac{a-x}{a+x}}, \sqrt{\frac{a+x}{a-x}}$

  Substitution : $x=a\cos 2\theta$ (or) $x=a\cos\theta$

- **Form of integrand :** $\sqrt{\frac{x}{a-x}}, \sqrt{\frac{a-x}{x}}$

  Substitution : $x=a\sin^2\theta$ (or) $x=a\cos^2\theta$

- **Form of integrand :** $\sqrt{\frac{x}{a+x}}, \sqrt{\frac{a+x}{x}}$

  Substitution : $x=a\tan^2\theta$ (or) $x=a\cot^2\theta$

- **Form of integrand :** $\sqrt{\frac{x-a}{b-x}}, \sqrt{(x-a)(b-x)}$

  $, \frac{1}{\sqrt{(x-a)(b-x)}}$

  Substitution : $x=a\cos^2\theta+b\sin^2\theta$

- **Form of integrand :** $\sqrt{\frac{x-a}{x-b}}, \sqrt{(x-a)(x-b)}$

  Substitution : $x=a\sec^2\theta-b\tan^2\theta$

- **Form of integrand :** $\frac{1}{\sqrt{(x-a)(x-b)}}$

  Substitution : $x-a=t^2$ or $x-b=t^2$

Important methods of Integration :

- If the given integral is of the form

$$\int\sqrt{ax^2+bx+c}\;dx \quad \text{(or)} \quad \int\frac{1}{\sqrt{ax^2+bx+c}}\,dx$$

(or) $\int\frac{1}{ax^2+bx+c}\,dx$ then replace ax^2+bx+c

by $\dfrac{\left[\frac{d}{dx}(ax^2+bx+c)\right]^2-\Delta}{4a}$

where $\Delta=b^2-4ac$

W.E-1 : ***Evaluate*** $\int\sqrt{2x^2+3x+4}\,dx$

Sol: Replace $2x^2+3x+4$ by $\frac{(4x+3)^2-23}{8}$

$$=\int\sqrt{\frac{(4x+3)^2-23}{8}}\,dx$$

$$=\frac{1}{\sqrt{8}}\left[\frac{4x+3}{2}\sqrt{(4x+3)^2-23}-\frac{23}{2}\cosh^{-1}\left(\frac{4x+3}{\sqrt{23}}\right)\right]\times\frac{1}{4}+C$$

W.E-2 : ***Evaluate*** $I=\int\frac{dx}{x^2+4x+7}$

Sol: Replace x^2+4x+7 by $\frac{(2x+4)^2+12}{4}$

$$I=\int\frac{4}{(2x+4)^2+12}\,dx=4\times\frac{1}{2}\times\frac{1}{\sqrt{12}}Tan^{-1}\left(\frac{2x+4}{\sqrt{12}}\right)$$

$$=\frac{1}{\sqrt{3}}Tan^{-1}\left(\frac{x+2}{\sqrt{3}}\right)+C$$

- $\int\frac{px+q}{ax^2+bx+c}dx=A\log\left|ax^2+bx+c\right|$

  $+B\int\frac{1}{ax^2+bx+c}dx$

- $\int(px+q)\sqrt{ax^2+bx+c}\,dx=$

  $\frac{2}{3}\mathrm{A}(ax^2+bx+c)^{3/2}+\mathrm{B}\int\sqrt{ax^2+bx+c}$ dx,

- $\int\frac{px+q}{\sqrt{ax^2+bx+c}}dx=$

  $2A\sqrt{ax^2+bx+c}+B\int\frac{1}{\sqrt{ax^2+bx+c}}\,dx$

  where $\mathrm{A}=\frac{p}{2a}$, $B=q-\frac{pb}{2a}$

  (or) Use $px+q=k\frac{d}{dx}\left(ax^2+bx+c\right)+l$

- . If the given integral is of the form

  $\int\frac{1}{(px+q)\sqrt{ax^2+bx+c}}\,dx$ (or)

$$\int \frac{1}{(px+q)^r\sqrt{ax^2+bx+c}}dx \quad \text{(or)}$$

$$\int \frac{x}{(px+q)\sqrt{ax^2+bx+c}}dx$$

then put $px+q=\frac{1}{t}$

➢ If the given integral is of the form

$$\int \frac{\sqrt{px+q}}{ax+b}dx, \ \int (ax+b)(\sqrt{px+q})dx,$$

$$\int \frac{1}{(ax+b)\sqrt{px+q}}dx$$

$$\int \frac{(ax+b)}{\sqrt{px+q}}dx \text{ then put } px+q=t^2$$

➢ If the given integral is of the form

$$\int \frac{1}{(ax^2+bx+c)\sqrt{px+q}}dx \text{ then put } px+q=t^2$$

➢ If the given integral is of the form

$$\int \frac{1}{a+b\sin x}dx \ \text{(or)} \int \frac{1}{a+b\cos x}dx \ \text{(or)}$$

$$\int \frac{1}{a\cos x+b\sin x+c}dx \text{ then put } \tan\left(\frac{x}{2}\right)=t$$

$$\Rightarrow dx=\frac{2dt}{1+t^2}, \ \sin x=\frac{2t}{1+t^2}, \ \cos x=\frac{1-t^2}{1+t^2}$$

➢ If the given integral is of the form

$$\int \frac{1}{a+b\sin 2x}dx \ \text{(or)} \int \frac{1}{a+b\cos 2x}dx \ \text{(or)}$$

$$\int \frac{1}{a\cos 2x+b\sin 2x+c}dx \text{ then put } \tan x=t$$

$$\Rightarrow dx=\frac{dt}{1+t^2}, \ \sin 2x=\frac{2t}{1+t^2}, \ \cos 2x=\frac{1-t^2}{1+t^2}$$

➢ $\int \frac{a\text{Cos}x+b\text{Sin}x}{c\text{Cos}x+d\text{Sin}x}dx=$

$$\left(\frac{ac+bd}{c^2+d^2}\right)x+\left(\frac{ad-bc}{c^2+d^2}\right)\log|\text{denominator}|+\lambda$$

W.E-3: Evaluate $\int \frac{dx}{1+\tan x}$

Sol: $\int \frac{\cos x}{\cos x+\sin x}dx$; $\quad a=1, b=0, c=1, d=1$

$$I=\frac{x}{2}+\frac{1}{2}\log|\cos x+\sin x|+C$$

➢ $\int \frac{aCosx+bSinx+l}{cCosx+dSinx+k}dx=$

$$\left(\frac{ac+bd}{c^2+d^2}\right)x+\left(\frac{ad-bc}{c^2+d^2}\right)\log|\text{denominator}|$$

$$+(l-Ak)\int \frac{1}{cCosx+dSinx+k}dx$$

where $A=\frac{ac+bd}{c^2+d^2}$

W.E-4: ***Evaluate*** $\int \frac{3\cos x+2\sin x+4}{4\cos x+3\sin x+5}dx$

Sol: $a=3, b=2, l=4, \quad c=4, d=3, k=5$

$$I=\frac{18}{25}x+\frac{1}{25}\log|4\cos x+3\sin x+5|+\frac{2}{5}\int \frac{1}{3\sin x+4\cos x+5}dx$$

$$I=\frac{18}{25}x+\frac{1}{25}\log|4\cos x+3\sin x+5|-\frac{4}{5\left(Tan\frac{x}{2}+3\right)}+C$$

➢ $I=\int \frac{ae^x+be^{-x}}{ce^x+de^{-x}}dx$, then take

$Nr=A(Dr)+B\frac{d}{dx}(Dr)$ then

$$I=\frac{1}{2}\left(\frac{a}{c}+\frac{b}{d}\right)x+\frac{1}{2}\left(\frac{a}{c}-\frac{b}{d}\right)\log\left|ce^{-x}+de^{-x}\right|+c$$

➢ If the given integral is of the form

$$\int \frac{1}{aCos^2x+bSin^2x+cCosx\sin x}dx \ \text{(or)}$$

$$\int \frac{1}{a+b\,Sin^2x}dx \ \text{(or)} \int \frac{1}{a+b\,Cos^2x}dx, \ (or)$$

$$\int \frac{1}{aCos^2x+bsin^2x+c}dx, (or) \quad \int \frac{1}{(aCosx+bSinx)^2}dx$$

then divide both numerator & denominator with $\cos^2 x$ and then take $\tan x=t$

W.E-5 : ***Evaluate*** $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx$

Sol: multiply & divided Nr with $\sec^2 x$

$$= \int \frac{\sec^2 x}{a^2 \tan^2 x + b^2} dx \; ; \; a \tan x = t$$

$$\sec^2 x \, dx = \frac{dt}{a} \; ; = \frac{1}{ab} \tan^{-1}\left(\frac{a \tan x}{b}\right) + C$$

- If the given integral is of the form $\int \frac{b + a \sin x}{(a + b \sin x)^n} dx$ then divide both numerator & denominator with $\cos^2 x$ and then take $a \sec x + b \tan x = t$
- If the given integral is of the form $\int \frac{b + a \cos x}{(a + b \cos x)^n} dx$ then divide both numerator & denominator with $\sin^2 x$ and then take $a \operatorname{cosec} x + b \cot x = t$

W.E-6 : $\int \frac{2 \sin x + 5}{(2 + 5 \sin x)^2} dx$

Sol: $= \int \frac{2 \sec x \tan x + 5 \sec^2 x}{(2 \sec x + 5 \tan x)^2} dx$

put $2 \sec x + 5 \tan x = t$

$$I = \int \frac{dt}{t^2} = -\frac{\cos x}{2 + 5 \sin x} + C$$

- If the given integral is of the form $\int \sin^m x \cos^n x \, dx$ then

(i) Put $\cos x = t$ if 'm' is odd and 'n' is even integer or n is a rational number.

(ii) Put $\sin x = t$ if 'n' is odd and 'm' is even integer or m is a rational number.

(iii) If both m,n are odd and m > n then take $\sin x = t$

(iv) If both m, n are rational numbers such that $m + n =$ negative even integer then put $\tan x = t$ (or) $\cot x = t$

- $$\int \frac{dx}{(ax^2 + bx + c)\sqrt{px^2 + qx + r}}$$

Case – I :- when $ax^2 + bx + c$ breaks up into two linear factors e.g.,

$$I = \int \frac{dx}{(x^2 - x - 2)\sqrt{x^2 + x + 1}} \text{ then}$$

$$\int \left(\frac{A}{x-2} + \frac{B}{x+1}\right) \frac{1}{\sqrt{x^2 + x + 1}} dx$$

$$A \int \frac{1}{(x-2)\sqrt{x^2 + x + 1}} dx +$$

$$B \int \frac{1}{(x+1)\sqrt{x^2 + x + 1}} dx$$

Put $x - 2 = \frac{1}{t}$

put $x + 1 = \frac{1}{t}$

Case – 2 :- If $ax^2 + bx + c$ is a perfect square say $(lx + my)^2$ then $lx + my = \frac{1}{t}$

Case – 3 :- If $b = 0$, $q = 0$,

e.g. $$\int \frac{dx}{(ax^2 + c)\sqrt{px^2 + r}}$$

then put $x = \frac{1}{t}$

- If the given integral is of the form

$$\int \frac{x^2 \pm 1}{x^4 + \lambda x^2 + 1} dx \text{ (or) } \int \frac{x^2}{x^4 + \lambda x^2 + 1} dx$$

(or) $$\int \frac{1}{x^4 + \lambda x^2 + 1} dx$$

where λ is a constant then divide both Nr,Dr with x^2 and then take $x + \frac{1}{x} = t$ (or) $x - \frac{1}{x} = t$

W.E - 7 : ***Evaluate*** $\int \frac{x^2-1}{x^4+1}dx$

Sol: divided Nr and Dr with x^2

$$\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx \quad \text{add on sub Dr by '2'}$$

$$=\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}+2-2}dx=\int \frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)^2-\left(\sqrt{2}\right)^2}dx$$

$$x+\frac{1}{x}=t \;;\; \left(1-\frac{1}{x^2}\right)dx=dt$$

$$=\int \frac{dt}{t^2-\left(\sqrt{2}\right)^2}=\frac{1}{2\sqrt{2}}\log\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right|$$

$$=\frac{1}{2\sqrt{2}}\log\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right|$$

➢ $\int \frac{1}{x^4+ax^2+b^2}dx=\frac{1}{2b}\int \frac{(x^2+b)-(x^2-b)}{x^4+ax^2+b^2}dx$ where b is a positive constant,

➢ $\int \frac{x^2}{x^4+ax^2+b^2}dx=\frac{1}{2}\int \frac{(x^2+b)+(x^2-b)}{x^4+ax^2+b^2}dx$ and split

➢ If the given integral is of the form $\int \frac{dx}{(x+a)^m(x+b)^n}$ when $m+n=2$, then the above integral becomes $\int\left(\frac{x+b}{x+a}\right)^m\frac{dx}{(x+b)^2}$ now take $\frac{x+a}{x+b}=t$

➢ If the given integral is of the form $\int \frac{dx}{(a-x)^m(b-x)^n}$ where $m+n=2$, then the above integral becomes $\int\left(\frac{b-x}{a-x}\right)^m\frac{dx}{(b-x)^2}$ now take $\frac{a-x}{b-x}=t$

➢ $\int(\tan x+\sec x)^n\sec^2 x\,dx=\frac{1}{2}\int(z^n+z^{n-2})dz$

By putting $z=\sec x+\tan x$

$$\therefore dz=\sec x(\sec x+\tan x)\Rightarrow \frac{dz}{z}=\sec x\;dx$$

$$\frac{1}{z}=\sec x-\tan x,\; z+\frac{1}{z}=2\sec x$$

W.E - 8 : ***Evaluate*** $\int(\tan x+\sec x)^2\sec^2 dx$

Sol: $n=2\;;=\frac{1}{2}\int\left(z^2+1\right)dz\;=\frac{1}{6}z^3+\frac{1}{2}z+C$

$$=\frac{1}{6}(\sec x+\tan x)^3+\frac{1}{2}(\sec x+\tan x)+C$$

➢ (i) If the given integral is of the form $\int R(x.x^{m/n},.....x^{r/s})dx$ where R is a rational function of its arguments, then put $x=t^k$,where $k=l.c.m$ of the denominators $\{n,.......s\}$

W.E - 9 : $I=\int \frac{dx}{(1+x)^{1/2}-(1+x)^{1/3}}$

Sol : $l.c.m\;\;of\;\;2,3\;is\;6$; put $1+x=t^6$

$$\therefore dx=6t^5dt\;;\quad I=6\int\frac{t^3}{(t-1)}dt$$

$$=6\left[\frac{t^3}{3}+\frac{t^2}{2}+t+\log|t-1|\right]+C\;\;where\;t=(1+x)^{1/6}$$

➢ $\int R\left(x,\left(\frac{ax+b}{cx+d}\right)^{1/n}\right)dx$, then put $\frac{ax+b}{cx+d}=t^n$

➢ If the given integral is of the form

$$\int\frac{1}{x\left(x^n+1\right)}dx,\;\int\frac{1}{x^n\left(x^n+1\right)^{1/n}}dx,$$

$\int\frac{1}{x^2\left(x^n+1\right)^{\frac{n-1}{n}}}dx$, then take x^n common in Dr.

and put $1+\frac{1}{x^n}=t$.

Integration By Using Partial Fractions:

- $\int \frac{1}{(ax+b)(cx+d)}dx = \frac{1}{ad-bc}\log\left|\frac{ax+b}{cx+d}\right| + K$
- $\int \frac{1}{(x^2+a^2)(x^2+b^2)}dx =$
  $\frac{1}{b^2-a^2}\left[\frac{1}{a}\tan^{-1}\frac{x}{a} - \frac{1}{b}\tan^{-1}\frac{x}{b}\right]+c$
- $\int \frac{x}{(x^2+a^2)(x^2+b^2)}dx = \frac{1}{2(a^2-b^2)}\log\left|\frac{x^2+b^2}{x^2+a^2}\right| + c$
- $\int \frac{1}{x(x^n+k)}dx = \frac{1}{n\,k}\log\left|\frac{x^n}{x^n+k}\right| + c$
- $\int \frac{1}{x(x^n-k)}dx = \frac{1}{nk}\log\left|\frac{x^n-k}{x^n}\right| + c$
- $\int \frac{1}{x(k-x^n)}dx = \frac{1}{n\,k}\log\left|\frac{x^n}{k-x^n}\right| + c$

Integral Of Inverse Trigonometric Functions:

- $\int Sin^{-1}x.dx = xSin^{-1}x + \sqrt{1-x^2} + c$
- $\int Cos^{-1}x.dx = xCos^{-1}x - \sqrt{1-x^2} + c$
- $\int Tan^{-1}x.dx = xTan^{-1}x - \frac{1}{2}\log(1+x^2) + c$
- $\int Cot^{-1}x.dx = xCot^{-1}x + \frac{1}{2}\log(1+x^2) + c$
- $\int Sec^{-1}x.dx = xSec^{-1}x - Cosh^{-1}x + c$
- $\int Co\sec^{-1}x.dx = xCo\sec^{-1}x + Cosh^{-1}x + c$
- $\int Sinh^{-1}x.dx = xSinh^{-1}x - \sqrt{x^2+1} + c$
- $\int Cosh^{-1}x\,dx = x\cos h^{-1}x - \sqrt{x^2-1} + c$
- $\int Sinh^{-1}\left(\frac{x}{a}\right).dx = xSinh^{-1}\left(\frac{x}{a}\right) - \sqrt{x^2+a^2} + c$
- $\int Cosh^{-1}\left(\frac{x}{a}\right)\cdot dx = xCosh^{-1}\left(\frac{x}{a}\right) - \sqrt{x^2-a^2} + c$

Reduction Formulae:

- For n $\in$ N, if $I_n = \int x^n e^{ax}dx$, then
  $I_n = \frac{e^{ax}}{a}\cdot x^n - \frac{n}{a}I_{n-1}$ for $n \ge 1$
- For n $\in$ N, if $I_n = \int Sin^n x\,dx$, then
  $I_n = \frac{-Sin^{n-1}xCosx}{n} + \frac{n-1}{n}I_{n-2}$ for $n \ge 2$
- For n $\in$ N, if $I_n = \int Cos^n x\,dx$, then
  $I_n = \frac{Cos^{n-1}xSinx}{n} + \frac{n-1}{n}I_{n-2}$ for $n \ge 2$
- For n $\in$ N, if $I_n = \int \tan^n x\,dx$, then
  $I_n = \frac{\tan^{n-1}x}{n-1} - I_{n-2}$ for $n \ge 2$
- For n $\in$ N, if $I_n = \int \cot^n x\,dx$, then
  $I_n = \frac{-\cot^{n-1}x}{n-1} - I_{n-2}$ for $n \ge 2$
- For n $\in$ N, if $I_n = \int Sec^n x\,dx$, then
  $I_n = \frac{Sec^{n-2}x\tan x}{n-1} + \frac{n-2}{n-1}\cdot I_{n-2}$ for $n \ge 2$
- For n $\in$ N, if $I_n = \int Co\sec^n x\,dx$, then
  $I_n = \frac{-\cos ec^{n-2}x\cot x}{n-1} + \frac{n-2}{n-1}\cdot I_{n-2}$ for $n \ge 2$
- For n $\in$ N, if $I_{m,n} = \int Sin^m xCos^n x\,dx$, (for $n \ge 2$)
  then $I_{m,n} = -\frac{Sin^{m-1}xCos^{n+1}x}{m+n} + \frac{m-1}{m+n}\cdot I_{m-2,n}$
  for $m \ge 2$
- For n $\in$ N, If $I_{n-1} = \int (\log x)^n\,dx$, then
  $I_n = x(\log x)^n - n.\,I_{n-1}$ *for* $n \ge 1$
- If $I_n = \int \frac{1}{(x^2+a^2)^n}dx$ $(n \in N)$ then
  $I_n = \frac{x}{(2n-2)a^2(x^2+a^2)^{n-1}}$
  $+\left(\frac{2n-3}{2n-2}\right)\frac{1}{a^2}I_{n-1}$ *for* $n \ge 2$

➢ If $I_n = \int \frac{t^n}{t^2+1} dt \left(n \in N\right)$ then

$$I_{n+2} = \frac{t^{n+1}}{n+1} - I_n$$

Note :- Replacing n by $n-2$ in above reduction formula we get $I_n = \frac{t^{n-1}}{n-1} - I_{n-2}$

➢ **To evaluate the integrals of the form**

i) $\int f(\sin 2x)(\sin x + \cos x)\,dx$

put $\sin x - \cos x = t$

ii) $\int f(\sin 2x)(\sin x - \cos x)\,dx$

put $\sin x + \cos x = t$

➢ **To evaluate integrals of the form**

i) $\int f\left(x+\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)dx$, put $x+\frac{1}{x}=t$

ii) $\int f\left(x-\frac{1}{x}\right)\left(1+\frac{1}{x^2}\right)dx$, put $x-\frac{1}{x}=t$

iii) $\int f\left(x^2+\frac{1}{x^2}\right)\left(x-\frac{1}{x^3}\right)dx$ put $x^2+\frac{1}{x^2}=t$

iv) $\int f\left(x^2-\frac{1}{x^2}\right)\left(x+\frac{1}{x^3}\right)dx$, put $x^2-\frac{1}{x^2}=t$

LEVEL -I (CW)

PROBLEMS BASED ON STANDARD INTEGRALS

1. $\int \frac{\cos 2x}{\cos^2 x . Sin^2 x} dx =$

1) - Tan x - cot x + c 2) Tan x + Sin x + c
3) Tan x + cot x + c 4) Tan x + Cos x + c

2. $\int \frac{1-Sin\,x}{1+Sin\,x} dx =$

1) 2 tan x - 2 Sec x - x + c
2) 2 tan x - Sec$x$ - x + c
3) tan x + 2 Sec x + x + c
4) tan x - 2 Sec x + x + c

3. $\int \frac{\cos 2x - \cos 2\alpha}{\sin x - \sin \alpha} dx =$

1) 2[cos x + (sin α)x] + c
2) 2[cos x - x sin α] + c
3) -2[cos x - x sin α] + c
4) -2[cos x - sin α] + c

4. $If\ x \in (-\pi,\pi),\ then\ \int \sqrt{1+\sin\frac{x}{2}}\,dx =$

1) $2\left[Sin\frac{x}{4} - Cos\frac{x}{4}\right]+c$ 2) $4\left[Sin\frac{x}{4} + Cos\frac{x}{4}\right]+c$
3) $4\left[Sin\frac{x}{4} - Cos\frac{x}{2}\right]+c$ 4) $4\left[Sin\frac{x}{4} - Cos\frac{x}{4}\right]+c$

5. $\int \frac{Cot^2 x}{(Co\sec^2 x + Co\sec x)} dx =$

1) x-Sin x + C 2) x + Cos x + C
3) sin x - x + C 4) x - cos x + C

6. **If** $\int \frac{1}{x+x^5} dx = f(x)+c$, **then** $\int \frac{x^4}{x+x^5} dx =$

1) $\log|x| + f(x) + c_1$ 2) $\log|x| - f(x) + k$
3) $x\, f(x) + c_1$ 4) $x \log|x| + c_1$

7. $\int \frac{1-\sqrt{x}}{1+\sqrt[4]{x}} dx =$

1) $x - \frac{4}{5}x^{5/4} + c$ 2) $x + \frac{4}{5}x^{5/4} + c$
3) $x - \frac{2}{5}x^{5/4} + c$ 4) $x - \frac{3}{5}x^{5/4} + c$

8. $\int Cot^{-1}(Sec\,x - Tan x)$ **dx = ...**

1) $\frac{\pi x}{4} - \frac{x^2}{4} + c$ 2) $\frac{\pi x}{4} + \frac{x^2}{4} + c$
3) $\frac{x^2}{4} + c$ 4) $\frac{x^2}{2} + c$

9. $\int \frac{dx}{\sqrt{x+a} + \sqrt{x+b}} =$

1) $\frac{1}{(a-b)}\left[(x+a)\sqrt{x+a} - (x+b)\sqrt{x+b}\right]+c$
2) $\frac{2}{3(a-b)}\left[(x+a)\sqrt{x+a} + (x+b)\sqrt{x+b}\right]+c$
3) $\frac{2}{3(a-b)}\left[(x+a)\sqrt{x+a} - (x+b)\sqrt{x+b}\right]+c$
4) $\sqrt{x+a} - \sqrt{x+b} + c$

10. $\int \{1+\tan x.\tan(x+\alpha)\}dx=$

1) $\cot\alpha \log|\cot(x+\alpha)|+c$

2) $\cot\alpha \log\left|\dfrac{\cot(x+\alpha)}{\cos x}\right|+c$

3) $\cot\alpha \log\left|\dfrac{\cos x}{\cos(x+\alpha)}\right|+c$

4) $\cot\alpha \log\left|\dfrac{\sin x}{\cot(x+\alpha)}\right|+c$

11. $\int \dfrac{\cos x-\sin x}{\cos x+\sin x}(2+2\sin 2x)dx$ **is equal to**

1) $\sin 2x+c$ 2) $\cos 2x+c$
3) $\tan 2x+c$ 4) $\cot 2x+c$

12. $\int \dfrac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}}dx=$

1) $\dfrac{1}{2\sqrt{1+x}}+c$ 2) $\dfrac{2}{3}(1+x)^{3/2}+C$

3) $\sqrt{1+x}+c$ 4) $2(1+x)^{3/2}+c$

13. If $\int \dfrac{x\sqrt{x}-1}{\sqrt{x}-1}dx = ax^2+bx^{3/2}+cx+d$ **then ascending order of a, b, c is**

1) a,b,c 2) b,c,a 3) c,a,b 4) d,b,a

14. $\int \dfrac{e^x}{e^{x/2}-1}dx=$

1) $2[e^{x/2}-1+\log|(e^{x/2}-1)|]+c$
2) $[e^{x/2}-1+\log|(e^{x/2}-1)|]+c$
3) $2[e^{x/2}+1+\log|(e^{x/2}+1)|]+c$
4) $[e^{x/2}+1+\log|(e^{x/2}-1)|]+c$

15. $\int \dfrac{x+(\arccos 3x)^2}{\sqrt{1-9x^2}}dx=$

1) $-\dfrac{1}{9}\left[\sqrt{1-9x^2}+(\arccos 3x)^3\right]+c$

2) $\sqrt{1+9x^2}+(\arccos 3x)^3+c$

3) $\dfrac{1}{9}\left[\sqrt{1-9x^2}+(\arccos 3x)^3\right]+c$

4) $\sqrt{1-9x^2}+(\arccos 3x)^3+c$

16. $\int 2^{-x}\tanh 2^{1-x}\,dx=$

1) $\log(\cosh 2^{1-x})+c$ 2) $\dfrac{-1}{2\log 2}\log(Cosh 2^{1-x})+c$

3) $(2\log 2)\log(\cosh 2^{1-x})+c$

4) $\dfrac{-1}{\log 2}\log(\cosh 2^{1-x})+c$

17. $\int 3^{3^{3^x}}3^{3^x}3^x\,dx=$

1) $\dfrac{3^{3^x}}{(\log 3)^3}+c$ 2) $\dfrac{3^{3^{3^x}}}{(\log 3)^3}+c$

3) $3^{3^x}(\log 3)^3+c$ 4) $3^{3^{3^x}}(\log 3)^3+c$

18. $\int \sin^5 x\cdot\cos^{100}x\,dx=$

1) $\dfrac{\cos^{105}x}{105}+2\dfrac{\cos^{103}x}{103}+\dfrac{\cos^{101}x}{101}+c$

2) $-\dfrac{\cos^{105}x}{105}+2\dfrac{\cos^{103}x}{103}-\dfrac{\cos^{101}x}{101}+c$

3) $-\dfrac{\cos^{105}x}{105}-2\dfrac{\cos^{103}x}{103}+\dfrac{\cos^{101}x}{101}+c$

4) $\dfrac{\cos^{105}x}{105}-2\dfrac{\cos^{103}x}{103}+\dfrac{\cos^{101}x}{101}+c$

INTEGRATION BY THE METHOD OF SUBSTITUTION

19.

$\int\left(\dfrac{f(x)\;g^1(x)-f^1(x)\;g(x)}{f(x)\;g(x)}\right).\left[\log\;(g(x))-\log(f(x))\right]\;dx=$

1) $\log\dfrac{g(x)}{f(x)}+c$ 2) $\dfrac{1}{2}\left(\log\dfrac{g(x)}{f(x)}\right)^2+c$

3) $\dfrac{g(x)}{f(x)}\log\left(\dfrac{g(x)}{f(x)}\right)+c$ 4) $\dfrac{f(x)}{g(x)}\log\left(\dfrac{g(x)}{f(x)}\right)+c$

20. $\int \dfrac{1}{1+Sin\,ax}dx=$

1) $\dfrac{1}{a}[Tan\,ax-Sec\,ax]+c$

2) $\dfrac{1}{a}[Co\sec ax-Cot\,ax]+c$

3) $\dfrac{1}{a}[Sec\,ax-Cot\,ax]+c$ 4) $2Tan(\dfrac{ax}{2})+c$

21. $\int \frac{1}{x\sqrt{a^2x^2-b^2}}dx =$

1) $\frac{1}{b}\sec^{-1}x+c$ 2) $\frac{1}{b}\sec^{-1}(ax)+c$

3) $\frac{1}{b}\sec^{-1}\left(\frac{ax}{b}\right)+c$ 4) $\frac{-1}{b}\sec^{-1}\left(\frac{ax}{b}\right)+c$

22. $\int \frac{10x^9+10^x\log_e 10}{10^x+x^{10}}dx =$

1) $\log|10^x+x^{10}|+c$ 2) $-\log|10^x+x^{10}|+c$
3) $10^x+x^{10}+c$ 4) $\log|10^x+x^9|+c$

23. $\int \frac{x^2}{\sqrt{1-x^6}}$ **dx =**

1) $\frac{1}{3}\mathrm{Sin}^{-1}(x^3)+c$ 2) $\frac{1}{3}\mathrm{Cos}^{-1}(x^3)+c$

3) $-\frac{1}{3}\mathrm{Sin}^{-1}(x^3)+c$ 4) $\frac{1}{4}\mathrm{Cos}^{-1}(x^3)+c$

24. $\int \frac{1}{x}\sqrt{\frac{x-1}{x+1}}dx =$

1) $\cosh^{-1}x-\sec^{-1}x+c$ 2) $\cosh^{-1}x+\sec^{-1}x+c$
3) $\sinh^{-1}x-\sec^{-1}x+c$ 4) $\sinh^{-1}x-\operatorname{cosec}^{-1}x+c$

25. If $f(x)=\cos x-\cos^2 x+\cos^3 x-....to\,\infty$ **then** $\int f(x)dx$ **is equal to**

1) $\tan\frac{x}{2}+c$ 2) $x-\tan\frac{x}{2}+c$

3) $x-\frac{1}{2}\tan\frac{x}{2}+c$ 4) $\frac{x-\tan\frac{x}{2}}{3}+c$

26. $\int \frac{\cos x}{\left[\cos\left(\frac{x}{2}\right)+\sin\left(\frac{x}{2}\right)\right]^5}dx =$

1) $\frac{2}{3}\left[\operatorname{cosec}\left(\frac{x}{2}\right)+\sec\left(\frac{x}{2}\right)\right]^3+C$

2) $\frac{2}{3}\left[\cos\left(\frac{x}{2}\right)+\sin\left(\frac{x}{2}\right)\right]^3+C$

3) $\frac{2}{3}\left[\sin\left(\frac{x}{2}\right)-\cos\left(\frac{x}{2}\right)\right]^{-3}+C$

4) $-\frac{2}{3}\left[\sin\left(\frac{x}{2}\right)+\cos\left(\frac{x}{2}\right)\right]^{-3}+C$

27. If $\frac{dI}{dy}=3^{\cos y}.\sin y$ **then I is equal to**

1) $3^{\cos y}+c$ 2) $-\frac{3^{\cos y}}{\log 3}+c$

3) $\sin y+c$ 4) $3^{\sin y}+c$

28. $\int 3^{-\log_9 x^2}dx =$

1) $2\log|x|+c$ 2) $\log|x|+c$
3) $-\log|x|+c$ 4) $-2\log|x|+c$

29. $\int \frac{a+b\sin x}{(b+a\sin x)^2}dx$ **, equals**

1) $\frac{-\cos x}{b+a\sin x}+c$ 2) $\frac{\cos x}{b+a\sin x}+c$

3) $\frac{\sin x}{b+a\sin x}+c$ 4) $\frac{-\sin x}{b+a\sin x}+c$

30. $\int \frac{e^{x-1}+x^{e-1}}{e^x+x^e}dx =$

1) $\log|e^x+x^e|+c$ 2) $\frac{1}{e}\log|e^x+x^e|+c$

3) $-\frac{1}{e}\log|e^x+x^e|+c$ 4) $-\log|e^x+x^e|+c$

31. $\int \frac{1}{\sin x\sqrt{\sin x\cos x}}dx =$

1) $\sqrt{\tan x}+c$ 2) $2\sqrt{\tan x}+c$
3) $-2\sqrt{\cot x}+c$ 4) $2\sqrt{\cot x}+c$

32. $\int \frac{\sqrt{1+\sqrt{x}}}{\sqrt{x}}dx =$

1) $4\sqrt{(1+x)^3}+c$ 2) $\frac{4}{3}\sqrt{(1+\sqrt{x})^3}+c$

3) $\frac{3}{4}\sqrt{(1+\sqrt{x})^3}+c$ 4) $\frac{4}{3}\sqrt{(1+\sqrt{x})^2}+c$

33. If $\int \frac{Sin\,2x}{a^2Cos^2x+b^2\sin^2 x}dx =$ $k.\log|a^2\cos^2 x+b^2\sin^2 x|+c$ **, then k =**

1) $\frac{1}{b^2-a^2}$ 2) $\frac{1}{(b^2-a^2)^2}$

3) $\frac{1}{a^2-b^2}$ 4) $\frac{1}{a^2+b^2}$

34. $\int e^{xSecx}\cdot \sec x(1+xTanx)dx=$

1) $e^{xSecx}+c$ 2) $-e^{xSecx}+c$

3) $\frac{1}{e^{xSecx}}+c$ 4) $-e^{x\tan x}+c$

35. $\int \frac{x}{\sqrt{1-x^2}.\cos^2\sqrt{1-x^2}}dx=$

1) $\tan\sqrt{1-x^2}+c$

2) $-\tan\sqrt{1-x^2}+c$

3) $\cot\sqrt{1-x^2}+c$

4) $-\cot\sqrt{1-x^2}+c$

36. **If** $\int \frac{e^x-1}{e^x+1}dx=f(x)+c, then f(x)=$

1) $2\log\left(e^x+1\right)$ 2) $\log\left(e^{2x}-1\right)$

3) $2\log\left(e^x+1\right)-x$ 4) $\log\left(e^{2x}+1\right)$

37. $If\int\sqrt{x}\left(1-x^3\right)^{-1/2}dx=\frac{2}{3}g\left(f\left(x\right)\right)+c\quad then$

1) $f\left(x\right)=\sqrt{x}, g\left(x\right)=\sin^{-1}x$

2) $f\left(x\right)=x^{3/2}, g\left(x\right)=\sin^{-1}x$

3) $f\left(x\right)=x^{2/3}, g\left(x\right)=\cos^{-1}x$

4) $f(x)=\sqrt{x}, g(x)=\cos^{-1}x$

38. $\int\left(\frac{\tan\frac{1}{x}}{x}\right)^2 dx=$

1) $x-\tan x+c$

2) $\frac{1}{x}-\tan\left(\frac{1}{x}\right)+c$

3) $\frac{1}{x}+\tan\left(\frac{1}{x}\right)+c$

4) $x+\tan x+c$

39. **The value** $\sqrt{2}\int\frac{\sin x\,dx}{\sin\left(x-\frac{\pi}{4}\right)}$ *is* **[AIE-2008]**

1) $x-\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+C$

2) $x+\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+C$

3) $x-\log\left|\cos\left(x-\frac{\pi}{4}\right)\right|+C$

4) $x+\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+C$

40. $\int(1-\cos x)\cos ec^2 x dx=f\left(x\right)+c\Rightarrow f\left(x\right)=$ **[EAMCET-2010]**

1) $Tan\frac{x}{2}$ 2) $Cot\frac{x}{2}$ 3) $2Tan\frac{x}{2}$ 4) $\frac{1}{2}Tan\frac{x}{2}$

41. $\int\frac{e^{2x}dx}{\sqrt[4]{e^x-1}}$ **is equal to**

1) $\frac{4}{21}\left(e^x-1\right)^{3/4}\left(3e^x+4\right)+k$

2) $\frac{\left(e^x-1\right)^{1/4}\left(3e^x+4\right)}{21}+k$

3) $\frac{4}{21}\left(e^x+1\right)\left(3e^x+4\right)+k$ 4) $(e^x-1)^{3/4}+k$

42. $\int\sin(\tan^{-1}x)\,dx=$

1) $\frac{1}{\sqrt{1+x^2}}+c$ 2) $\frac{1}{\sqrt{1-x^2}}+c$

3) $\sqrt{1+x^2}+c$ 4) $\sqrt{1-x^2}+c$

43. $\int Tan^3 2x\sec 2x\,dx=$

1) $Sec^3 2x+3\sec 2x+c$

2) $\frac{1}{6}(Sec^3 2x-3\sec 2x)+c$

3) $Sec^3 2x-3\sec 2x+c$

4) $\frac{-1}{6}(Sec^3 2x-3\sec 2x)+c$

44. $\int \frac{e^x(1+x)}{Sin^2(xe^x)}dx=$

1) $\tan(xe^x)+c$ 2) $-\cot(xe^x)+c$
3) $\sin(xe^x)+c$ 4) $\sin e^x+c$

45. $\int xe^{2x}(1+x)dx$, **equal to**

1) $\frac{xe^x}{2}+c$ 2) $\frac{(e^x)^2}{2}+c$ 3) $\frac{(1+x)^2}{2}+c$ 4) $\frac{(e^x x)^2}{2}+c$

46. **If** $\int \frac{\cos x-\sin x}{\sqrt{8-\sin 2x}}dx=\sin^{-1}\left(\frac{\sin x+\cos x}{a}\right)+c$ **then a=**

1) 2 2) 3 3) 4 4) 5

47. *If* $\int f(x)dx=f(x)$, *then* $\int\{f(x)\}^2dx$ **is equal to**

1) $\frac{1}{2}\{f(x)\}^2$ 2) $\{f(x)\}^3$ 3) $\frac{(f(x))^3}{3}$ 4) $\{f(x)\}^2$

48. $\int \frac{(x-x^5)^{1/5}}{x^6}dx$ **is equal to**

1) $\frac{5}{24}\left(\frac{1}{x^4}-1\right)^{6/5}+c$ 2) $\frac{5}{24}\left(1-\frac{1}{x^4}\right)^{6/5}+c$

3) $\frac{-5}{24}\left(\frac{1}{x^4}-1\right)^{6/5}+c$ 4) $\frac{5}{24}\left(1+\frac{1}{x^4}\right)^{6/5}+c$

49. $\int \frac{1}{Tan\,x\,Tan(60^0-x)Tan(60^0+x)}dx=$

1) $\frac{1}{3}\log|\sin 3x|+c$ 2) $3\log|\sin 3x|+c$

3) $\frac{1}{2}\log|\sin 3x|+c$ 4) $\frac{1}{3}\log|\cos 3x|+c$

50. $\int \frac{1+x\sin x+\cos x}{x(1+\cos x)}dx=$

1) $\log|x|+\log\left|Sec\frac{x}{2}\right|+c$

2) $\log|x|+2\log\left|Sec\frac{x}{2}\right|+c$

3) $\log|x|-2\log\left|Sec\frac{x}{2}\right|+c$

4) $\log|\sec x-\tan x|+c$

51. **If** $\frac{\pi}{4}<x<\frac{\pi}{2}$, **then**

$\int \frac{\sin x-\cos x}{\sqrt{1-\sin 2x}}e^{\sin x}\cos x\,dx=$

1) $e^{\sin x}+c$ 2) $e^{\sin x-\cos x}+c$

3) $e^{\sin x+\cos x}+c$ 4) $e^{\cos x-\sin x}+c$

52. **If** $\int \frac{dx}{ae^{mx}+be^{-mx}}=KTan^{-1}(Pe^{mx})+C$ **,thenK, P=**

1) $K=\frac{1}{\sqrt{ab}},P=\sqrt{\frac{a}{b}}$

2) $K=\frac{1}{m\sqrt{ab}},P=\sqrt{\frac{a}{b}}$

3) $K=m\sqrt{ab},P=\sqrt{\frac{b}{a}}$ 4) $K=\frac{1}{m\sqrt{ab}},P=\sqrt{\frac{b}{a}}$

53. $\int \frac{(1-\cos x)dx}{\cos x(1+\cos x)}=$

1) $\log(\sec x+\tan x)-2\tan\frac{x}{2}+c$

2) $\log(\sec x+\tan x)+2\tan\frac{x}{2}+c$

3) $\log(\sec x-\tan x)-2\tan\frac{x}{2}+c$

4) $\log(\sec x-\tan x)+2\tan\frac{x}{2}+c$

INTEGRATION AS THE INVERSE PROCESS OF DIFFERENTIATION

54. **If** $\int(x^2+2x^4+3x^6)(1+x^2+x^4)^{1/2}dx$

$=k(Ax^2+Bx^4+cx^6)^p+l$ **then**

1) $k=\frac{1}{3},A=B=C=p$

2) $k=\frac{1}{3},A=B=C,p=3/2$

3) $k=3,p=1/3,A=B=C$

4) $k=-\frac{1}{3},A=B=C,P=-\frac{3}{2}$

55. If the primitive of $\frac{1}{f(x)}$ is equal to $\log\{f(x)\}^2 + c$, then $f(x)$ is

1) $x+d$ 2) $\frac{x}{2}+d$ 3) $\frac{x^2}{2}+d$ 4) x^2+d

56. g(x) is an antiderivative of f(x) = $1+2^x\log 2$ whose graph passes through $(-1,\frac{1}{2})$. The curve y = g(x) meets y - axis at

1) (0,1) 2) (0,2)
3) (0, -2) 4) (1, 1)

57. The equation of a curve passing through origin is given by $y=\int x^3\cos x^4 dx$. If the equaiton of the curve is written in the form $x=g(y)$, then

1) $g(y)=\sqrt[3]{\sin^{-1}(4y)}$

2) $g(y)=\sqrt{\sin^{-1}(4y)}$

3) $g(y)=\sqrt[4]{\sin^{-1}(4y)}$

4) $g(y)=\sqrt[5]{\sin^{-1}(4y)}$

58. The function $f(x)$ whose graph passes through the point $\left(0,\frac{7}{3}\right)$ and whose derivative is $x\sqrt{1-x^2}$ is given by.

1) $-\frac{1}{3}\left[\left(1-x^2\right)^{3/2}-8\right]$ 2) $-\frac{1}{3}\left[\left(1+x^2\right)^{3/2}-8\right]$

3) $-\frac{1}{3}\left[\left(x^2-1\right)^{3/2}-8\right]$ 4) $-\frac{1}{3}\left[\left(1-x^2\right)^{3/2}+7\right]$

59. $\int\frac{\sin^2 x.\sec^2 x+2\tan x.\sin^{-1}x.\sqrt{1-x^2}}{\sqrt{1-x^2}\left(1+\tan^2 x\right)}dx=$

1) $\left(\cos^2 x\right).\left(\sin^{-1}x\right)+c$

2) $\left(\sin^2 x\right).\left(\sin^{-1}x\right)+c$

3) $\left(\sec^2 x\right).\left(\cos^{-1}x\right)+c$

4) $\left(\sec^2 x\right).\left(\tan^{-1}x\right)+c$

INTEGRATION BY PARTS

60. If $f(x)$ is a polynomial of nth degree then $\int e^x f(x)dx=$ Where $f^n(x)$ denotes nth order derivative of $f(x)$ w.r.t. x

1) $e^x\left[f(x)-f^{1}(x)+f^{11}(x)-f^{111}(x)+......+(-1)^n f^n(x)\right]$

2) $e^x\left[f(x)+f^{1}(x)+f^{11}(x)+f^{111}(x)+......+(-1)^n f^n(x)\right]$

3) $e^x\left[f(x)+f^{1}(x)+f^{11}(x)+f^{111}(x)+......+(-1)^n f^{2n}(x)\right]$

4) $e^x\left[f(x)+f^{1}(x)+f^{11}(x)+f^{111}(x)+......+(-1)^n f^{3n}(x)\right]$

61. $\int e^x\left(\frac{x^4+x^2+1}{x^2+x+1}\right)dx=$

1) $e^x(x^4+x^2+1)+c$ 2) $e^x(x^2+x+1)+c$

3) $e^x(x^2-3x+4)+c$

4) $e^x(x^2-4x+5)+c$

62. $\int x^2 2^{3x}dx=$

1) $\frac{2^{3x}x^2}{3\log 2}-\frac{x.2^{3x}}{9.(\log 2)^2}+\frac{2^{3x}}{27(\log 2)^3}$+c

2) $\frac{x^2 2^{3x}}{3\log 2}-\frac{x.2^{3x+1}}{9.(\log 2)}+\frac{2^{3x+1}}{27(\log 2)}+c$

3) $\frac{x^2 2^{3x}}{3\log 2}-\frac{2^{3x+1}.x}{3^2(\log 2)^2}+\frac{2^{3x+1}}{3^3(\log 2)^3}+c$

4) $\frac{x^2 2^{3x}}{\log 2}-\frac{2^{3x+1}.x}{3^2(\log 2)^2}+\frac{2^{3x+1}}{(\log 2)^3}+c$

63. $\int x\left\{f\left(x^2\right)g''\left(x^2\right)-f''\left(x^2\right)g\left(x^2\right)\right\}dx$ is equal to

1) $f\left(x^2\right)g'\left(x^2\right)-g\left(x^2\right)f'\left(x^2\right)+c$

2) $\frac{1}{2}\left\{f\left(x^2\right)g\left(x^2\right)f'\left(x^2\right)\right\}+c$

3) $\frac{1}{2}\left\{f\left(x^2\right)g'\left(x^2\right)-g\left(x^2\right)f'\left(x^2\right)\right\}+c$

4) $f(x^2)g(x^2)+c$

64. $\int Cos\,x \log(Cosx)\,dx =$ **Sinx log(Cosx) +log|Sec x + Tan x| +f(x) + c, then f(x) =**
1) -Sin x 2) Cos x
3) Tan x 4) Cot x

65. $\int x^3.Cos\,x^2\,dx =$
1) $x^2 \text{Sin}\, x^2 - \text{Cos}\, x^2 + c$
2) $\frac{1}{2}[x^2 \text{Sin}\, x^2 + \text{Cos}\, x^2] + c$
3) $\frac{1}{3}[x^2 \text{Sin}\, x^2 + \text{Cos}\, x^2] + c$
4) $x^2 \text{Sin}\, x^2 + \text{Cos}\, x^2 + c$

66. $\int \exp\sqrt[3]{x}\,dx =$
1) $x^{2/3} - 2x^{1/3} + 2 + c$
2) $(x^{2/3} - 2x^{1/3} + 2)\exp(\sqrt[3]{x}) + c$
3) $3(x^{2/3} - 2x^{1/3} + 2)\exp(\sqrt[3]{x}) + c$
4) $2(x^{2/3} - 2x^{1/3} + 2)\exp(\sqrt[3]{x}) + c$

67. $\int x\tan x\sec^2 x\,dx =$
1) $\frac{1}{2}[x\tan^2 x - \tan x + x] + c$
2) $\frac{1}{2}[x\tan^2 x - \tan x - x] + c$
3) $\frac{1}{2}[x\tan^2 x + \tan x - x] + c$
4) $x\tan^2 x - \tan x + x + c$

68. $\int Sin\sqrt{x}\,dx =$
1) $2(\text{Sin}\sqrt{x} - \sqrt{x}.\cos\sqrt{x}) + c$
2) $2(\text{Sin}\sqrt{x} + \sqrt{x}.\cos\sqrt{x}) + c$
3) $2(\text{Sin}\sqrt{x} - \frac{1}{2}\sqrt{x}\cos\sqrt{x}) + c$
4) $2(\text{Sin}\sqrt{x} + \frac{1}{2}\sqrt{x}\cos\sqrt{x}) + c$

69. $\int (\log x)^2\,dx =$
1) $x[(\log x)^2 - 2\log x + 2] + c$
2) $x[(\log x)^2 + 2\log x + 2] + c$
3) $[(\log x)^2 - 2\log x + 2] + c$
4) $[(\log x)^2 + 2\log x + 2] + c$

PROBLEMS BASED ON $\int e^x[f(x)+f^1(x)]dx$

70. $If \int e^x(f(x)-f^1(x))dx = \phi(x),$

$then \int e^x f(x)dx =$
1) $\phi(x)+e^x f(x)$ 2) $\phi(x)-e^x f(x)$
3) $\frac{1}{2}\{\phi(x)+e^x f(x)\}$ 4) $\frac{1}{2}\{\phi(x)-e^x f^1(x)\}$

71. $\int e^x\left[\frac{x+4}{(x+6)^3}\right]dx =$
1) $\frac{e^x}{(x+6)^2}+c$ 2) $e^x\frac{1}{(x+4)^2}+c$
3) $e^x\frac{x}{x+6}+c$ 4) $e^x\frac{4}{(x+6)^2}+c$

72. $\int e^x\left(\frac{1-Sin\,x}{1-Cos\,x}\right)$ **dx =** **(E-2008)**
1) $e^x \text{Tan}\frac{x}{2} + c$ 2) $-e^x \text{Cot}\frac{x}{2} + c$
3) $e^x \csc^2\frac{x}{2} + c$ 4) $e^{-x}\cot\frac{x}{2} + c$

73. $\int 2e^x\frac{\sqrt{1+Sin2x}}{1+Cos2x}\,dx=$
1) e^x Secx + c 2) e^x Tan x + c
3) e^x Secx Tan x + c 4) e^x cosec x + c

74. $\int e^x\frac{x^2+1}{(x+1)^2}dx =$
1) $e^x\left(\frac{x-1}{x+1}\right)+c$ 2) $e^x\left(\frac{x+1}{x-1}\right)+c$
3) $e^x\frac{x-1}{(x+1)^2}+c$ 4) $e^x\frac{x+1}{(x-1)^2}+c$

75. $\int e^x\left[\frac{x^3+x+1}{(1+x^2)^{3/2}}\right]$ **dx =**
1) $e^x\frac{x}{\sqrt{1+x^2}}+c$ 2) $e^x.\frac{e^x}{\sqrt{1+x^2}}+c$
3) $e^x+\frac{x}{\sqrt{1+x^2}}+c$ 4) $e^x\sqrt{1+x^2}+c$

76. $\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx$ **is equal to**

1) $\frac{x}{(\log x)^2+1}+c$ 2) $\frac{xe^x}{1+x^2}+c$

3) $\frac{x}{x^2+1}+c$ 4) $\frac{\log x}{(\log x)^2+1}+c$

77. $\int e^{x/2}\sec 2x(1+4\tan 2x)dx$ **is equal to**

1) $4e^{x/2}\sec 2x+c$ 2) $2e^{x/2}\sec 2x+c$

3) $e^{x/2}\sec 2x+c$ 4) $\frac{1}{2}e^{x/2}\sec 2x+c$

78. $\int e^x(\log x+\frac{1}{x^2})dx=$

1) $e^x\log x+c$ 2) $e^x(\log x-\frac{1}{x})+c$

3) $e^x(\log x+\frac{1}{x})+c$ 4) $\frac{e^x}{x^2}+c$

79. $\int\frac{(\log x)^2-\log x+1}{\left[(\log x)^2+1\right]^{3/2}}dx=$

1) $\frac{x}{\left[(\log x^2+1)\right]^{3/2}}+c$ 2) $\frac{x+1}{\sqrt{(\log x)^2+1}}+c$

3) $\frac{x}{\sqrt{(\log x)^2+1}}+c$ 4) $\frac{x}{\left[(\log x)^2+1\right]^{3/2}}+c$

$\int\left[f(x)+xf^1(x)\right]dx$ **MODEL**

80. $\int\frac{x+\sin x}{1+Cos\,x}$ **dx =**

1) $x\tan\frac{x}{2}+c$ 2) $x\cot\frac{x}{2}+c$

3) $x\sin\frac{x}{2}+c$ 4) $x\cos\frac{x}{2}+c$

81. $\int(1+x-x^{-1})e^{x+x^{-1}}dx=$

1) $(x+1)e^{x+x^{-1}}+c$ 2) $(x-1)e^{x+x^{-1}}+c$

3) $-xe^{x+x^{-1}}+c$ 4) $xe^{x+x^{-1}}+c$

82. $\int e^{\frac{x}{2}}Sin\left(\frac{\pi}{4}+\frac{x}{2}\right)dx=$

1) $\sqrt{2}\,e^{\frac{x}{2}}Sin\,\frac{x}{2}+c$

2) $\sqrt{2}\,e^{\frac{x}{2}}Cos\frac{x}{2}+c$

3) $-\sqrt{2}\,e^{x/2}Sin\frac{x}{2}+c$

4) $-\sqrt{2}\,e^{x/2}Cos\frac{x}{2}+c$

PROBLEMS BASED ON

$\int e^{ax}\sin(bx)dx$ *and* $\int e^{ax}\cos(bx)dx$

83. $\int 3^x\,Cos\,4x\,dx=$

1) $\frac{3^x}{(\log 3)^2+16}[(\log 3)Cos\,4x+4\,Sin\,4x]+c$

2) $\frac{3^x}{9+(\log 4)^2}[Cos\,4x+(\log 3)\cdot Sin\,4x]+c$

3) $\frac{3^x}{(\log 3)^2+(\log 4)^2}[(\log 3)Sin\,4x+(\log 4)Cos\,3x]+c$

4) $\frac{3^x}{(\log 3)^2+16}[(\log 3)Sin4x+(\log 4)Cos3x]+c$

PROBLEMS ON INVERSE TRIGONOMETRIC FUNCTIONS

84 $\int Sin^{-1}\left(\frac{2x}{1+x^2}\right)dx=f(x)-\log(1+x^2)+c$

$\Rightarrow f(x)=$

1) $2x\,Tan^{-1}x$ 2) $-2x\,Tan^{-1}x$
3) $x\,Tan^{-1}x$ 4) $-x\,Tan^{-1}x$

85. $\int\frac{Sin^{-1}x-Cos^{-1}x}{Sin^{-1}x+Cos^{-1}x}dx=$

1) $\frac{4}{\pi}\left[x\,Sin^{-1}x+\sqrt{1-x^2}\right]-x+c$

2) $\log\left[Sin^{-1}x+Cos^{-1}x\right]+c$

3) $\frac{4}{\pi}\left[x\,Sin^{-1}x+\sqrt{1-x^2}\right]+c$

4) $\frac{4}{\pi}\left[x\,Sin^{-1}x-\sqrt{1-x^2}\right]+c$

APPLICATIONS OF STANDARD INTEGRALS

86. $\int \frac{ax^2-b}{x\sqrt{c^2x^2-(ax^2+b)^2}}dx=$

1) $Sin^{-1}\left(\frac{ax+b/x}{c}\right)+k$

2) $Sin^{-1}\left(\frac{ax^2+b/x^2}{c}\right)+k$

3) $Cos^{-1}\left(\frac{ax^2+b/x^2}{c}\right)+k$

4) $Cos^{-1}\left(\frac{ax^2-b/x^2}{c}\right)+k$

87. **If** $\int \frac{1}{\sqrt{2ax-x^2}}dx=(f0g)(x)+c$, **then**

1) $f(x)=Sin^{-1}x,\ g(x)=\frac{x+a}{a}$

2) $f(x)=Sin^{-1}x,\ g(x)=\frac{x-a}{a}$

3) $f(x)=Cos^{-1}x,\ g(x)=\frac{x-a}{a}$

4) $f(x)=Tan^{-1}x,\ g(x)=\frac{x-a}{a}$

88. $If \int \frac{1}{Cos^2x(1-4\tan^2x)}dx$

$=k.\log\left|\frac{1+2\tan x}{1-2\tan x}\right|+c$, **then k =**

1) 1 2) 3 3) 1/3 4) 1/4

89. $\int x^2\sqrt{x^6-1}dx=$

1) $\frac{1}{6}\left\{x^3\sqrt{x^6-1}-\log\left(x^3+\sqrt{x^6-1}\right)\right\}+c$

2) $\frac{1}{6}\left\{x^3\sqrt{x^6-1}+\log\left(x^3+\sqrt{x^6-1}\right)\right\}+c$

3) $\frac{1}{6}\left\{x^3\sqrt{x^6-1}-\sin^{-1}\left(x^3\right)\right\}+c$

4) $\frac{1}{6}\left\{x^3\sqrt{x^6-1}+\sinh^{-1}\left(x^3\right)\right\}+c$

METHODS OF INTEGRATION

90. $\int \frac{x^2+1}{x^4+1}dx=$

1) $\frac{1}{\sqrt{2}}Tan^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)+c$ 2) $Tan^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)+c$

3) $\frac{1}{\sqrt{2}}\mathrm{Tan}^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+c$ 4) $Tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+c$

91. $\int \frac{x^4+1}{1+x^6}dx=$

1) $\mathrm{Tan}^{-1}(x)-\mathrm{Tan}^{-1}(x^3)+c$

2) $\mathrm{Tan}^{-1}(x)-\frac{1}{3}\mathrm{Tan}^{-1}(x^3)+c$

3) $\mathrm{Tan}^{-1}(x)+\mathrm{Tan}^{-1}(x^3)+c$

4) $\mathrm{Tan}^{-1}(x)+\frac{1}{3}\mathrm{Tan}^{-1}(x^3)+c$

92. **If** $\int \frac{2x+5}{\sqrt{x^2-2x+10}}$ **dx**

$=2\sqrt{x^2-2x+10}+7\,\mathrm{Sinh}^{-1}f(x)+c$, **then f(x) =**

1) x -1 2) 3(x - 1) 3) $\frac{(x-1)}{3}$ 4) $\frac{\sqrt{x-1}}{3}$

93. $\int \frac{e^x}{\sqrt{5-4e^x-e^{2x}}}dx=$

1) $\cos^{-1}\left(\frac{e^x+2}{3}\right)+c$ 2) $\cos^{-1}\left(\frac{e^x-3}{2}\right)+c$

3) $\sin^{-1}\left(\frac{e^x+2}{3}\right)+c$ 4) $\sin^{-1}\left(\frac{e^x-3}{2}\right)+c$

94. **If** $\int \frac{1}{\sqrt{x^2+x+1}}dx=$ **a** $\sinh^{-1}$ **(bx+c)+d ,**

then descending order of a, b, c is

1) a, b, c 2) b, c, a 3) b, a, c 4) c, a, b

95. $\int \frac{dx}{(x+100)\sqrt{x+99}}=f(x)+c \Rightarrow f(x)=$

1) $2(x+100)^{\frac{1}{2}}$ 2) $3(x+100)^{\frac{1}{2}}$

3) $2Tan^{-1}\sqrt{x+99}$ 4) $2Tan^{-1}\sqrt{x+100}$

96. $\int \frac{x+1}{\sqrt{2x+1}}dx=$

1) $\sqrt{2x+1}\cdot\left(\frac{x+2}{3}\right)+c$ 2) $\sqrt{2x+1}.(x+2)+c$

3) $\frac{3}{\sqrt{2x+1}(x+2)}+c$ 4) $\frac{(2x+1)\sqrt{x+2}}{3}+c$

97. $\int \frac{dx}{7+5\cos x}=$

1) $\frac{1}{\sqrt{3}}Tan^{-1}\left(\frac{1}{\sqrt{3}}Tan\frac{x}{2}\right)+c$

2) $\frac{1}{\sqrt{6}}Tan^{-1}\left(\frac{1}{\sqrt{6}}Tan\frac{x}{2}\right)+c$

3) $\frac{1}{7}Tan^{-1}\left(Tan\frac{x}{2}\right)+c$ 4) $\frac{1}{4}Tan^{-1}\left(Tan\frac{x}{2}\right)+c$

98. $\int \frac{dx}{1-\sin x-\cos x}=$

1) $\log\left|1+Tan\frac{x}{2}\right|+c$ 2) $\log\left|1-Tan\frac{x}{2}\right|+c$

3) $\log\left|1-\cot\frac{x}{2}\right|+c$ 4) $\log\left|1+\cot\frac{x}{2}\right|+c$

99. $\int \frac{\sin x}{\sin x-\cos x}dx=$

1) $\frac{1}{2}\left[(x-\log|(\sin x-\cos x|\right]+c$

2) $\frac{1}{2}\left[(x-\log|(\sin x+\cos x|\right]+c$

3) $\frac{1}{2}\left[(x+\log|(\sin x-\cos x|\right]+c$

4) $\frac{1}{2}\left[\log|(\sin x-\cos x|\right]+c$

100. If $\int \frac{9\cos x-\sin x}{4\sin x+5\cos x}dx=$

$A\log|4\sin x+5\cos x|+Bx+c$, then A=...,B=...,

1) $\frac{1}{2},\frac{-1}{2}$ 2) 1, 1 3) - 1, 1 4) 1, -1

101. $\int \frac{dx}{9\sin^2 x+16\cos^2 x}=$

1) $Tan^{-1}\left(\frac{3\tan x}{4}\right)+c$ 2) $Tan^{-1}\left(\frac{4\tan x}{3}\right)+c$

3) $\frac{1}{12}Tan^{-1}\left(\frac{3\tan x}{4}\right)+c$ 4) $\frac{1}{12}Tan^{-1}\left(\frac{4\tan x}{3}\right)+c$

102. If $\int \frac{Sin\,x}{\sin^2 x+4\cos^2 x}dx=\frac{1}{\sqrt{3}}Tan^{-1}\left(\frac{g(x)}{\sqrt{3}}\right)+c$ **, then** $g(x)=$

1) Sec x 2) Tan x 3) Sin x 4) Cos x

INTEGRATION BY USING PARTIAL FRACTIONS

103. $\int \frac{1+x^5}{1+x}dx$ is equal to:

1) $1-x+x^2-x^3+x^4+c$

2) $x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}+c$

3) $(1+x)^5+c$ 4) $(1-x)^5+c$

104. $\int \frac{ax+b}{cx+d}dx=$

1) $\frac{ax}{c}-\frac{bc-ad}{c^2}\log|cx+d|+k$

2) $\frac{ax}{c}+\frac{bc-ad}{c^2}\log|cx+d|+k$

3) $\log\left|\frac{ax+b}{cx+d}\right|+k$ 4) $\frac{ax}{c}+\frac{b}{c}\log|cx+d|+k$

105. $\int \frac{x^2}{(a+bx)^2}dx=$

1) $\frac{1}{b^3}\left[(a+bx)-2a\log(a+bx)-\frac{a^2}{a+bx}\right]+c$

2) $\frac{1}{b^3}\left[2a\log(a+bx)+\frac{a^2}{a+bx}\right]+c$

3) $\frac{1}{b^3}\left[2a-(a+bx)^2+a^2\right]+c$

4) $\frac{1}{b^3}\left[2a\log(a+bx)-\frac{a^2}{a+bx}\right]+c$

106. $\int \frac{d\theta}{(Sin\,\theta-2\cos\theta)(2\sin\theta+\cos\theta)}=$

1) $\frac{1}{5}\log\left|\frac{\tan\theta-2}{2\tan\theta+1}\right|+c$ 2) $\frac{1}{2}\log\left|\frac{\tan\theta-2}{2\tan\theta+1}\right|+c$

3) $\frac{1}{5}\log\left|\frac{2\tan\theta+1}{\tan\theta-2}\right|+c$ 4) $\frac{1}{2}\log\left|\frac{2\tan\theta-1}{\tan\theta+2}\right|+c$

107. $\int \frac{dx}{Sin^2x+Sin2x}=$

1) $\frac{1}{2}\log\left|\frac{Tanx}{Tanx+2}\right|+c$ 2) $\frac{-1}{2}\log\left|\frac{Tanx}{Tanx+2}\right|+c$

3) $\frac{1}{3}\log\left|\frac{Tanx}{Tanx+2}\right|+c$ 4) $-\frac{1}{3}\log\left|\frac{Tanx}{Tanx+2}\right|+c$

108. $\int \frac{(x+1)}{x(1+xe^x)}dx=$

1) $\log\left|\frac{e^x}{1+xe^x}\right|+c$ 2) $-\log\left|\frac{e^x}{1+xe^x}\right|+c$

3) $\log\left|\frac{xe^x}{1+xe^x}\right|+c$ 4) $-\log\left|\frac{e^x}{1-xe^x}\right|+c$

109. $\int \frac{1}{\sin x\cos x(\tan^9 x+1)}dx=$

1) $\log\left|\frac{Cos^9x}{\sin^9 x+\cos^9 x}\right|+c$

2) $\frac{1}{9}\log\left|\frac{Sin^9x}{\sin^9 x+\cos^9 x}\right|+c$

3) $\frac{1}{9}\log\left|\frac{Sin^9x+\cos^9 x}{\sin^9 x}\right|+c$

4) $\frac{1}{9}\log\left|\frac{\cos^9 x}{\sin^9 x+\cos^9 x}\right|+c$

110. $\int \frac{\left(\sqrt{x}\right)^3}{\left(\sqrt{x}\right)^5+x^4}dx = A\log\left(\frac{x^k}{x^k+1}\right)+c$

then the values of A & K respectively are

1) $\frac{3}{2},\frac{2}{3}$ 2) $\frac{3}{2},2$ 3) 2, 3/2 4) $\frac{2}{3},\frac{3}{2}$

REDUCTION FORMULAE

111. $\int \tan^7 x\,dx = f(x)+\log|\cos x|+c$ **then** $f(x)=$

1) $\frac{1}{6}\tan^6 x+\frac{1}{4}\tan^4 x+\frac{1}{2}\tan^2 x$

2) $\frac{1}{6}\tan^6 x+\frac{1}{4}\tan^4 x-\frac{1}{2}\tan^2 x$

3) $\frac{1}{6}\tan^6 x-\frac{1}{4}\tan^4 x-\tan^2 x$

4) $\frac{1}{6}\tan^6 x-\frac{1}{4}\tan^4 x+\frac{1}{2}\tan^2 x$

112. $\int Co\sec^4 x\,dx=$

1) $Cot\,x+\frac{1}{3}Cot^3x+\frac{2x}{3}$+c

2) $Cot\,x-\frac{1}{3}Cot^3x+c$

3) $Cot\,x+\frac{1}{3}Cot^3x+c$ 4) $-Cot\,x-\frac{1}{3}Cot^3x+c$

113. If $I_n=\int\frac{t^n}{1+t^2}dt$ **then** $I_{n+2}=$

1) $\frac{t^{n+1}}{n+1}-I_n$ 2) $\frac{t^n}{n}-I_n$

3) $-\frac{t^{n+1}}{n+1}-I_n$ 4) $\frac{t^{n+2}}{n+2}-I_n$

114. If $I_n=\int\frac{\sin nx}{\sin x}dx\ (n>1)$ **then** $I_{n+1}-I_{n-1}=$

1) $\frac{2}{n-1}\cos(n-1)x$ 2) $\frac{2}{n-1}\sin(n-1)x$

3) $\frac{2}{n}\cos nx$ 4) $\frac{2}{n}\sin nx$

115. $I_n=\int\frac{Cos\,nx}{Cos\,x}dx$ **(n > 1) then** $I_n=$

1) $-\frac{2}{(n-1)}Cos(n-1)x+I_{n-2}$

2) $\frac{2}{(n-1)}Cos(n-1)x+I_{n-2}$

3) $\frac{2}{(n-1)}Sin(n-1)x-I_{n-2}$

4) $-\frac{2}{(n-1)}Sin(n-1)x+I_{n-2}$

116. $\int \frac{1}{\left(x^2-25\right)^{3/2}}dx=$

1) $\frac{1}{25}\frac{x}{\sqrt{x^2-25}}+c$ 2) $-\frac{1}{25}\frac{x}{\sqrt{x^2-25}}+c$

3) $\frac{1}{25}\frac{1}{\sqrt{x^2-25}}+c$ 4) $-\frac{1}{25}\frac{1}{\sqrt{x^2-25}}+c$

117. $\int \frac{\sin x+\cos x}{\sqrt{1+\sin 2x}}dx$ **is equal to**

1) $x+c$ for $\frac{3\pi}{4}<x<\frac{7\pi}{4}$ 2) $|x|+c, \forall x\in R$

3) $-x+c$ for $-\frac{\pi}{4}<x<\frac{3\pi}{4}$

4) $x+c$ for $\frac{-\pi}{4}<x<\frac{3\pi}{4}$

118. $\int 5^{\log_e x}dx$ **is equal to**

1) $\frac{x^{\log_e 5+1}}{\log_e 5+1}+k$ 2) $5^{\log_e x}.\log 5$

3) $e^{\log_5 x}+k$ 4) $\frac{5^{\log_e x}}{x}+k$

119. $\int |x+y|dx$ **where** $\frac{dy}{dx}=0$ **is equal to**

1) 0 2) $\frac{(x+y)^2}{2}+c$

3) $\frac{-(x+y)^2}{2}+c$ 4) $\frac{(x+y)|x+y|}{2}+c$

120. $\int \frac{d^2}{dx^2}\left(\tan^{-1}x\right)dx$ **is equal to**

1) $\frac{1}{1+x^2}+c$ 2) $\tan^{-1}x+c$

3) $x\tan^{-1}x-\frac{1}{2}\log\left(1+x^2\right)+c$

4) $\frac{1}{\sqrt{1+x^2}}+c$

121. The indefinite integral of $\sin x$ **w.r.to** $\cos x$ **is**

1) $\frac{\sin 2x}{4}+\frac{x}{2}+c$ 2) $\frac{\sin 2x}{4}-\frac{x}{2}+c$

3) $2\sin 2x+c$ 4) $\sin x+\cos x+c$

LEVEL-I (C.W.) - KEY

1) 1	2)1	3)2	4)4	5)2	6)2
7)1	8)2	9)3	10)3	11)1	12)2
13)1	14)1	15)1	16)2	17)2	18)2
19)2	20)1	21)3	22)1	23)1	24)1
25)2	26)4	27)2	28)2	29)1	30)2
31)3	32)2	33)1	34)1	35)2	36)3
37)2	38)2	39)2	40)1	41)1	42)3
43)2	44)2	45)4	46)2	47)1	48)3
49)1	50)2	51)1	52)2	53)1	54)2
55)2	56)2	57)3	58)1	59)2	60)1
61)3	62)3	63)3	64)1	65)2	66)3
67)1	68)1	69)1	70)3	71)1	72)2
73)1	74)1	75)1	76)1	77)2	78)2
79)3	80)1	81)4	82)1	83)1	84)1
85)1	86)1	87)2	88)4	89)1	90)3
91)4	92)3	93)3	94)3	95)3	96)3
97)2	98)3	99)3	100)2	101)3	102)1
103)2	104)2	105)1	106)1	107)1	108)3
109)2	110)4	111)4	112)4	113)1	114)4
115)3	116)2	117)4	118)1	119)4	120)1
121)2					

LEVEL-I (C.W.) - HINTS

1. $\cos 2x=\cos^2 x-\sin^2 x$
2. Multiply numerator and denominator by $1-\sin x$
3. $\cos 2x=1-2\sin^2 x,\ \cos 2\alpha=1-2\sin^2\alpha$
4. $\sqrt{1+\sin\frac{x}{2}}=\sin\frac{x}{4}+\cos\frac{x}{4}$
5. $\cot^2 x=\operatorname{cosec}^2 x-1$
6. $\frac{x^4}{x+x^5}=\frac{(x^4+1)-1}{x(x^4+1)}$
7. $1-\sqrt{x}=(1+\sqrt[4]{x})(1-\sqrt[4]{x})$
8. $\sec x-\tan x=\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)=\cot\left(\frac{\pi}{4}+\frac{x}{2}\right)$

9. Multiply numerator and denominator by $\sqrt{x+a}-\sqrt{x+b}$
10. $(x+\alpha)-x=\alpha$ .Taking tan on both sides
11. Write $1+\sin 2x=(\cos x+\sin x)^2$
12. $I=\int\sqrt{1+x}dx$
13. $\left(\sqrt{x}\right)^3-1=\left(\sqrt{x}-1\right)\left(x+\sqrt{x}+1\right)$
14. Write $e^x = e^{x/2} . e^{x/2}$ and then put $e^{x/2} -1 = t$
15. Put $\cos^{-1}3x=t$
16. Put $2^{1-x} = t$
17. Put $3^{3^x}=t$
18. Put $\cos x=t$
19. Put $\log\left[\frac{g(x)}{f(x)}\right]=t$
20. Multiply numerator and denominator by $1-\sin ax$
21. Put $ax=t$
22. Apply the formula $\int\frac{f^1(x)}{f(x)}dx$
23. Put $x^3=t$ and apply the formula $\int\frac{1}{\sqrt{1-x^2}}dx$
24. Under square root multiplying and dividing with $(x-1)$
25. $f(x)=\frac{\cos x}{1+\cos x}=\frac{(1+\cos x)-1}{1+\cos x}$
split and integrate
26. Put $\cos x=\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)$
27. Put $\cos y=t$ and then integrate
28. $\int 3^{-\log_9 x^2}dx=\int\frac{1}{x}dx$
29. Dividing Nr. and Dr. by $\cos^2 x$ and using the formula $\int\frac{f'(x)}{[f(x)]^2}dx=\frac{-1}{f(x)}+c$
30. Put $e^x+x^e=t$
31. Put $\cot x=t$
32. Put $1+\sqrt{x}=t$
33. Put $a^2\cos^2 x+b^2\sin^2 x=t$
34. Put $x\sec x=t$
35. Put $\sqrt{1-x^2}=t$
36. $\frac{e^x-1}{e^x+1}=\frac{(e^x+1)-2}{e^x+1}$
37. Put $x^{3/2}=t$
38. Put $\frac{1}{x}=t$
39. Put $x-\frac{\pi}{4}=t$
40. $G.I.=\frac{1-\cos x}{1-\cos^2 x}=\frac{1}{1+\cos x}$
41. Put $e^x-1=t^4$
42. Put $\tan^{-1}x=t$
43. Put $\sec 2x=t$
44. Put $xe^x=t$
45. Put $x.e^x=t$
46. Put $\sin x+\cos x=t$
47. Put $f(x)=e^x$
48. Put $\frac{1}{x^4}-1=t$
49. $\tan x\tan(60^0-x)\tan(60^0+x)=\tan 3x$
50. $I=\int\frac{(1+\cos x)+x\sin x}{x(1+\cos x)}dx$
51. $\sqrt{1-\sin 2x}=\sin x-\cos x$
52. $I=\int\frac{e^{mx}}{a\left(e^{mx}\right)^2+b}dx$
53. $1-\cos x=(1+\cos x)-2\cos x$
54. $(x^2+2x^4+3x^6)(1+x^2+x^4)^{1/2}=$
$(x+2x^3+3x^5)(x^2+x^4+x^6)^{1/2}$
and using $\int[f(x)]^n f'(x)dx$
55. $\int\frac{1}{f(x)}dx=\log\{f(x)\}^2+c$ differentiate on both sides
56. Given $g(x)=\int f(x)$ where $f(x)=1+2^x\log 2$

57. Put $x^4 = t$ $\therefore y = \frac{1}{4}\sin x^4 + c$.

58. $\frac{dy}{dx} = x\sqrt{1-x^2}$

$\therefore y = -\frac{1}{3}(1-x^2)^{3/2} + c$ passes through $\left(0, \frac{7}{3}\right)$

59. Divide nr with dr

60. Using by parts n times taking e^x as second function

61. $x^4 + x^2 + 1 = (x^2+x+1)(x^2-x+1)$

62. Put $U = x^2, V = 2^{3x}$

63. Put $x^2 = t$ and using byparts

64. Put $U = \log\cos x, V = \cos x$

65. Put $x^2 = t$

66. Put $x = t^3$

67. Put $U = x, \quad V = \tan x \sec^2 x$

68. $Put\sqrt{x} = t$

69. Put $\log x = t \Rightarrow x = e^t$

70. $\int e^x\left[f(x) - f'(x)\right]dx = \phi(x) \to (1)$

$\int e^x\left[f(x) + f'(x)\right]dx = e^x f(x) \to (2)$

Adding (1) and (2)

71. $x + 4 = (x+6) - 2$

72. $\frac{1-\sin x}{1-\cos x} = \frac{1}{2}\operatorname{cosec}^2\frac{x}{2} - \cot\frac{x}{2}$

73. $\sqrt{1+\sin 2x} = \cos x + \sin x$

74. Apply the formula $\int e^x\left[f(x) + f^1(x)\right]dx$

75. $x^3 + x + 1 = x(1+x^2) + 1$

76. Take $\log_e x = t \Rightarrow x = e^t$

77. Put $\frac{x}{2} = t$

78. $I = \int e^x\left(\log x + \frac{1}{x} - \frac{1}{x} + \frac{1}{x^2}\right)dx$

79. Put $\log x = t$

80. $\frac{x+\sin x}{1+\cos x} = \frac{x + 2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}$

81. $e^{x+\frac{1}{x}}\left[1 + x - \frac{1}{x}\right] = xe^{x+\frac{1}{x}}\left[\frac{1}{x} + \left(1 - \frac{1}{x^2}\right)\right]$

82. $I = \frac{1}{\sqrt{2}}\int e^{x/2}\left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)dx$ & put t =x/2

83. $I = \int\left(e^{\log 3}\right)^x \cos 4x\, dx$

84. $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1} x$

85. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

86. Put $ax + \frac{b}{x} = t$

87. $2ax - x^2 = a^2 - (x-a)^2$

88. Put $\tan x = t$

89. Put $x^3 = t$

90. Put $x - \frac{1}{x} = t$

91. multiplying and dividing with $(1+x^2)$

92. $\frac{2x+5}{\sqrt{x^2-2x+10}} = \frac{2x-2+7}{\sqrt{x^2-2x+10}} =$

$\frac{2x-2}{\sqrt{x^2-2x+10}} + \frac{7}{\sqrt{x^2-2x+10}}$

93. Put $e^x = t$ and $5 - 4t - t^2 = 9 - (t+2)^2$

94. $x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

95. Put $x + 99 = t^2$

96. Put $2x + 1 = t^2$

97. Put $\tan\frac{x}{2} = t$

98. Put $\tan\frac{x}{2} = t$

99. Use $\int\frac{a\cos x + b\sin x}{c\cos x + d\sin x}dx = \frac{ac+bd}{c^2+d^2}x$

$+\frac{ad-bc}{c^2+d^2}\log\left|c\cos x + d\sin x\right| + k$

100. Use $\int \frac{a\cos x+b\sin x}{c\cos x+d\sin x}dx=\frac{ac+bd}{c^2+d^2}x$
$+\frac{ad-bc}{c^2+d^2}\log\left|c\cos x+d\sin x\right|+c$

101. Multiply numerator and denominator with $\sec^2 x$ and put Tanx = t

102. Put $\sec x=t$

103. $\frac{1+x^5}{1+x}=x^4-x^3+x^2-x+1$

104. $\frac{ax+b}{cx+d}=\frac{a}{c}+\frac{bc-ad}{c}\frac{1}{cx+d}$

105. Put a + bx = t

106. Multiply Numerator & Denominator by $Cos^2\theta$

107. Divide Numerator & Denominator by $\cos^2 x$

108. Multiply Numerator & Denominator by e^x.

109. $\sin x\cos x(\tan^9 x+1)$
$=\cos^2 x\tan x(\tan^9 x+1)$ and put $\tan x=t$

110. $I=\int\frac{1}{(\sqrt{x})^2+(\sqrt{x})^5}dx=\int\frac{1}{x\left(x^{\frac{3}{2}}+1\right)}dx$

111. Apply the formula $I_n=\frac{\tan^{n-1}x}{n-1}-I_{n-2}$

112. Apply the formula
$I_n=\frac{-Co\sec^{n-2}x\,Cotx}{n-1}+\frac{n-2}{n-1}.I_{n-2}$

113. Put t = tan x

114. $\sin(n+1)x-\sin(n-1)x=2\cos nx\sin x$

115. $\cos nx+\cos(n-2)x=2\cos(n-1)x\cos x$

116. $x=5\sec\theta$

117. $\sqrt{1+\sin 2x}=\pm(\sin x+\cos x)$
$=\sin x+\cos x$ for $-\frac{\pi}{4}<x<\frac{3\pi}{4}$

118. $I=\int x^{\log_e 5}dx$

119. y is constant

120. $\int\frac{d}{dx}\left(f(x)\right)dx=f(x)+c$

121. $d(\cos x)=-\sin x$; $I=\int-\sin^2 x\,dx$

LEVEL - I (H.W)

PROBLEMS BASED ON STANDARD INTEGRALS

1. $\int\frac{1+\cos^2 x}{1+\cos 2x}dx=$

1) Tan x - x + c 2) cot x + x + c
3) $\frac{Tanx}{2}+x+c$ 4) $\frac{1}{2}(Tan\,x+x)+c$

2. $\int\frac{Sin^3x+\cos^3 x}{Sin^2x.\cos^2 x}dx=$

1) Sec x - Cosec x + c 2) sec x + cosec x + c
3) Sin x - Cos x + c 4) sin x + cos x + c

3. $\int\frac{e^x dx}{\cos hx+\sin hx}=$

1) log cos hx+c 2) tan x + cot x + c
3) $\frac{1}{2}e^{2x}+c$ 4) x + c

4. $\int\frac{\sqrt{1-x^2}+\sqrt{1+x^2}}{\sqrt{1-x^4}}dx=$

1) $\cosh^{-1}x+\sin^{-1}x+c$
2) $\cosh^{-1}x+\cos^{-1}x+c$
3) $\sinh^{-1}x+\sin^{-1}x+c$ 4) $\sinh^{-1}x+\cos^{-1}x+c$

5. If n is a positive odd integer, then $\int\left|x^n\right|dx=$

1) $\left|\frac{x^{n+1}}{n+1}\right|+c$ 2) $\frac{x^{n-1}}{n+1}+c$ 3) $\frac{\left|x^n\right|x}{n+1}+c$ 4) $\frac{x^{n+1}}{n+1}+c$

6. $\int\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-..........\right)dx=$

1) Sin x + c 2) -Sin x+c
3) Cos x+c 4) -Cos x + c

7. $\int\sqrt{Cos\,x}.Sin^3x\,dx=$

1) $\frac{-2}{3}Cos^{3/2}x+\frac{2}{7}Cos^{7/2}x+c$
2) $\frac{2}{3}Cos^{3/2}x+\frac{2}{7}Cos^{7/2}x+c$
3) $\frac{1}{3}Cos^{3/2}x-\frac{2}{7}Cos^{7/2}x+c$
4) $\frac{1}{3}Cos^{3/2}x+\frac{2}{7}Cos^{7/2}x+c$

INTEGRATION BY THE METHOD OF SUBSTITUTION

8. $\int x f^1(ax^2+b)\sqrt{f(ax^2+b)}dx=$

1) $\frac{\{f(ax^2+b)\}^{3/2}}{2}+c$ 2) $x\{f(ax^2+b)\}^2+c$

3) $\frac{\{f(ax^2+b)\}^{3/2}}{3a}+c$ 4) $\frac{\{f(ax^2+b)\}^2}{2a}+c$

9. $\int\frac{(\cos x)^{n-1}}{(\sin x)^{n+1}}dx=$

1) $\frac{-\cot^n x}{n}+c$ 2) $\frac{-\cot^n x}{n+1}+c$

3) $\frac{\cot^n x}{n}+c$ 4) $\frac{\cot^n x}{n+1}+c$

10. $\int\frac{Co\sec x}{\log\left|\tan\frac{x}{2}\right|}dx=$

1) $\log\left|\log\left(\tan\frac{x}{2}\right)\right|+c$ 2) $-\log\left|\log\left(\tan\frac{x}{2}\right)\right|+c$

3) $\log\left|\log\left(\cot\frac{x}{2}\right)\right|+c$

4) $-\log\left|\log\left(\cot\frac{x}{2}\right)\right|+c$

11. $\int Sec\,x\cdot\log(Sec\,x+\tan x)dx=$

1) $[\log(\sec x+\tan x)]^2+c$

2) $\frac{[\log(\sec x+\tan x)]^2}{2}+c$

3) $-\log(\sec x+\tan x)+c$

4) $\log(\sec x+\tan x)+c$

12. $\int\frac{1-\tan x}{1+\tan x}dx=$

1) $\log|1+\tan x|+c$ 2) $\log|1-\tan x|+c$

3) $\log|\sin x+\cos x|+c$

4) $\log|\sin x-\cos x|+c$

13. $\int\frac{\tan x-\tan\frac{x}{2}}{1+\tan x\tan\frac{x}{2}}dx=$

1) $\log\left|\sec\frac{x}{2}\right|+c$ 2) $2\log\left|\cos\frac{x}{2}\right|+c$

3) $2\log\left|\sec\frac{x}{2}\right|+c$ 4) $\log\left|\cos\frac{x}{2}\right|+c$

14. $\int\sec^p x.Tan\,xdx=$

1) $\frac{\sec^p x}{p}+c$ 2) $\frac{\sec^{p-1}x}{p-1}+c$

3) $\frac{\sec^{p+1}x}{p+1}+c$ 4) $\frac{\sec^{p-1}x}{p+1}+c$

15. $\int\frac{x\sin^{-1}x^2}{\sqrt{1-x^4}}dx=$

1) $\frac{Sin^{-1}x^2}{4}+c$ 2) $\frac{x}{\sqrt{1-x^4}}+c$

3) $\frac{-[Sin(x^2)]^2}{4}+c$ 4) $\frac{[Sin^{-1}(x^2)]^2}{4}+c$

16. $\int\frac{Sec\,x}{(Sec\,x+Tan\,x)^2}dx=$

1) $\frac{1}{2(Sec\,x+Tan\,x)^2}+c$

2) $\frac{-1}{2(Sec\,x+Tan\,x)^2}+c$

3) $\frac{1}{2(Sec\,x+Tan\,x)}+c$ 4) $\frac{-1}{2(Sec\,x+Tan\,x)}+c$

17. $\int x^6\sin(5x^7)dx=\frac{k}{5}\cos(5x^7)+c, then\,k=$

1) 7 2) -7 3) $1/7$ 4) $\frac{-1}{7}$

18. $\int x\cos^3 x^2.\sin x^2 dx=$

1) $\frac{1}{8}\sin^4 x^2+c$ 2) $\frac{1}{8}\cos^4 x^2+c$

3) $-\frac{1}{8}\cos^4 x^2+c$ 4) $-\frac{1}{8}\sin^4 x^2+c$

19. $\int \frac{Cosx}{1+\sin^2 x}dx=$

1) $Tan^{-1}(Sin^2 x)+c$ 2) $\log|1+Sin^2 x|+c$

3) $Tan^{-1}(Sin\,x)+c$ 4) $Tan^{-1}(\cos x)+c$

20. $\int Sec(\tan x).Sec^2 x\,dx=$

1) $\log\left|\tan\left(\frac{\pi}{4}+\frac{1}{2}\tan x\right)\right|+c$

2) $\log\left|\tan\left(\frac{\pi}{4}-\frac{1}{2}\tan x\right)\right|+c$

3) $\log|\sec x+\tan x|+c$

4) $\log|\sec x-\tan x|+c$

21. $\int \frac{e^{Tan^{-1}\sqrt{x}}}{\sqrt{x}+x\sqrt{x}}dx=$

1) $e^{Tan^{-1}\sqrt{x}}+c$ 2) $\frac{1}{2}e^{Tan^{-1}\sqrt{x}}+c$

3) $\log Tan^{-1}\sqrt{x}+c$ 4) $2e^{Tan^{-1}\sqrt{x}}+c$

22. $\int (x^x)^2(1+\log x)dx$

1) $\frac{(x^x)^2}{2}+k$ 2) x^x+k

3) $\frac{x^x}{2}+k$ 4) $\frac{x}{2}+c$

23. **If** $\int \frac{2^x}{\sqrt{1-4^x}}$ **dx = k Sin⁻¹2ˣ + c, then** $k=$

1) $\log 2$ 2) $\frac{1}{\log 2}$

3) $\frac{1}{2\log 2}$ 4) $2\log 2$

24. $\int \frac{\tan x+\sec x-1}{\tan x-\sec x+1}dx=$

1) $\log|\sec x+\tan x|+c$

2) $\log|\sec x-\tan x|+c$

3) $\log|\sec x(Sec\,x-Tan\,x)|+c$

4) $\log|Sec\,x(Sec\,x+Tan\,x|+c$

25. **I :** $\int \frac{1}{\sin x\cos^2 x}dx=\log\left|\tan\frac{x}{2}\right|+\sec x+c$

II : $\int \frac{\sin^3 x+\cos^3 x}{\sin^2 x.\cos^2 x}dx=\sec x+\operatorname{cosec} x+c$

1) only I is true 2) only II is true

3) both I and II are true 4) neither I nor II true

26. **If** $\int \frac{\cos 4x+1}{\cot x-\tan x}dx=A\cos 4x+B,$ **then :**

1) $A=-\frac{1}{2}$ 2) $A=-\frac{1}{8}$ 3) $A=-\frac{1}{4}$ 4) $A=-\frac{1}{6}$

27. $\int \frac{\cos x+x\sin x}{x(x+\cos x)}dx=$

1) $\log\left|\frac{x}{x+\cos x}\right|+c$ 2) $-\log\left|\frac{x}{x+\cos x}\right|+c$

3) $\log\left|\frac{x+\cos x}{2x}\right|+c$ 4) $-\log\left|\frac{x+\cos x}{2x}\right|+c$

28. $\int \frac{1}{x^2}e^{\frac{x-1}{x}}dx=$

1) $e^{\frac{x}{x-1}}+c$ 2) $e^{\frac{x-1}{x}}+c$ 3) $-e^{\frac{x}{x-1}}+c$ 4) $-e^{\frac{x-1}{x}}+c$

29. $\int \frac{dx}{\cos x\sqrt{Cos\,2x}}=$

1) $Cos^{-1}(Tan\,x)+c$ 2) $Cos(Tan^{-1}x)+c$

3) $Sin^{-1}(Tan\,x)+c$ 4) $Tan(Sin^{-1}x)+c$

30. $\int x\log x(\log x-1)dx$ **is equal to**

1) $2(x\log x-x)^2+c$ 2) $\frac{1}{2}(x\log x-x)^2+c$

3) $(x\log x)^2+c$ 4) $\frac{1}{2}(x\log x)^3+c$

31. $\int \sqrt{\frac{x}{a^3-x^3}}dx=$

1) $\frac{2}{3}\cos^{-1}\left(\frac{x}{a}\right)^{3/2}+c$ 2) $\frac{1}{3}Sin^{-1}\left(\frac{x}{a}\right)^{3/2}+c$

3) $\frac{2}{3}Sin^{-1}\left(\frac{x}{a}\right)^{3/2}+c$ 4) $\frac{1}{3}Cos^{-1}\left(\frac{x}{a}\right)^{3/2}+c$

32. $\int \frac{\sec^2 x}{\log(\tan x)^{\tan x}} dx =$

1) $\log|\tan x| + C$ 2) $\log|\log(\tan x)| + C$

3) $\frac{1}{\log|\tan x|} + C$ 4) $\frac{\tan x}{\log|\tan x|} + C$

33. The antiderivative of every odd function is

1) an odd function 2) an even function
3) neither even nor odd
4) sometimes even, sometimes odd

INTEGRATION BY PARTS

34. $\int x^2 e^x \, dx =$

1) $e^x(x^2 - 2x + 2) + c$ 2) $e^x(x^2 + 2x + 2) + c$
3) $x^2 + e^x + c$ 4) $e^x(x^2 + x + 2) + c$

35. $\int e^x 2^{3\log_2 x} dx = e^x f(x) + c$**, then f(x) =**

1) $x^3 - 3x^2 + 6x - 6$ 2) $x^3 - 3x^2 - 6x - 6$
3) $x^3 - 3x^2 + 6x + 6$ 4) $x^3 + 3x^2 + 6x + 6$

36. $\int \log_{10} x \, dx$ **is equal to**

1) $(x-1)\log_e x + c$

2) $\log_e 10 . x \log_e\left(\frac{x}{e}\right) + c$

3) $\log_{10} e . x \log_e\left(\frac{x}{e}\right) + c$ 4) $\frac{1}{x} + c$

37. $\int\left[\frac{Cosx}{x} - \sin x \ \ \log x\right] dx =$

1) $(\log x)\sin x + c$
2) $(\log x)(\text{Cos } x) + c$
3) $-(\log x)(\text{Cos } x) + c$
4) $-(\log x)\text{ Sin } x + c$

38. The antiderivative of the function $(3x+4)$ $|\sin x|$**, when** $0 < x < \pi$ **,is given by**

1) $3\sin x - (3x+4)\cos x + c$
2) $3\sin x + (3x+4)\cos x + c$
3) $-3\sin x + (3x+4)\cos x + c$
4) $3\cos x + (3x+4)\sin x + c$

39. $\int x^n . \log x \, dx =$

1) $\frac{x^{n+1}}{(n+1)^2}\left(\log x - \frac{1}{(n+1)^2}\right) + c$

2) $\left(\log x - \frac{1}{(n+1)^2}\right) + c$

3) $\frac{x^{n+1}}{(n+1)}\left(\log x - \frac{1}{(n+1)}\right) + c$

4) $\frac{x^{n+1}}{(n+1)}\left(\log x - \frac{1}{(n+1)^2}\right) + c$

40. $\int e^{Sin\,x} Sin\,2x \, dx =$

1) $2e^{Sin\,x}(Sin\,x + 1) + c$ 2) $e^{Sin\,x}(Sin\,x + 2) + c$
3) $e^{Sin\,x}(2Sin\,x - 2) + c$ 4) $e^{Sin\,x}(3Sin\,x - 2) + c$

41. $\int\left[(\log x)^2 - 7\log x + 9\right] dx =$

1) $x\left[(\log x)^2 - 9\log x\right] + c$
2) $x\left[(\log x)^2 - 9(\log x) + 18\right] + c$
3) $x\left[(\log x)^2 + 9\log x - 18\right] + c$
4) $x\left[(\log x)^2 - 7\log x + 9\right] + c$

PROBLEMS BASED ON

$$\int e^x\left[f(x) + f^1(x)\right]dx$$

42. $\int e^x \csc x(1 - \cot x)\, dx =$

1) $e^x \cot x + c$ 2) $e^x \csc x \cot x + c$
3) $e^x \csc x + c$ 4) $-e^x \cot x + c$

43. $\int e^x(1 + Tan\,x + Tan^2\,x)\, dx =$

1) $e^x \text{Tan } x + c$ 2) $e^x(1 + \text{Tan}^2 x) + c$
3) $e^x \text{Tan}^2 x + c$ 4) $e^x(1 + \text{Tan } x) + c$

44. $\int e^x x^x(2 + \log x)\, dx =$

1) $x^x + c$ 2) $e^x \log x + c$
3) $e^x x^x + c$ 4) $e^x + x^x + c$

45. $\int e^x\left(x + \sqrt{1+x^2}\right)\left(1 + \frac{1}{\sqrt{1+x^2}}\right)$ **dx =**

1) $e^x \cos h^{-1} x + c$ 2) $\frac{e^x}{\sqrt{1+x^2}} + c$

3) $e^x \text{Sin } h^{-1} x + c$ 4) $e^x\left(x + \sqrt{x^2+1}\right) + c$

46. $\int e^x \left[\log(x+\sqrt{x^2-1}) + \frac{d}{dx}(\cosh^{-1}x) \right] dx =$
1) $e^x \sinh^{-1} x + c$
2) $e^x \log(x+\sqrt{x^2+1}) + c$
3) $e^x \cosh^{-1} x + c$
4) $-e^x \sinh^{-1} x + c$

47. $\int \frac{\log(x/e)}{(\log x)^2} dx$
1) $\frac{\log x}{x}+c$ 2) $\frac{x}{\log x}+c$
3) $\frac{x}{(\log x)^2}+c$ 4) $\frac{(\log x)^2}{x}+c$

48. **The value of** $\int e^{2x}(2\sin 3x + 3\cos 3x)\,dx$ **is**
1) $e^{2x}\sin 3x + c$ 2) $e^{2x}\cos 3x + c$
3) $e^{2x}+c$ 4) $e^{2x}(2\sin 3x)+c$

49. $\int e^{-x}\csc x(\cot x+1)\,dx =$
1) $-e^{-x}\csc x + c$ 2) $e^{x}\csc x + c$
3) $-e^{x}\cot x + c$ 4) $-e^{-x}\cot x + c$

$\int [f(x)+xf^1(x)]dx$ **MODEL**

50. $\int e^{\sin^{-1}x}\left[1+\frac{x}{\sqrt{1-x^2}}\right] dx =$
1) $x\, e^{\sin^{-1}x} + c$ 2) $e^{\sin^{-1}x} + c$
3) $\frac{1}{\sqrt{1-x^2}} e^{\sin^{-1}x} + c$ 4) $x^2\, e^{\sin^{-1}x} + c$

51. $\int e^{\tan^{-1}x}\left[\frac{1+x+x^2}{1+x^2}\right] dx =$
1) $x^2 e^{\tan^{-1}x} + c$ 2) $x\, e^{\tan^{-1}x} + c$
3) $e^{\tan^{-1}x} + c$ 4) $\frac{1}{2} e^{\tan^{-1}x} + c$

PROBLEMS BASED ON $\int e^{ax}\sin(bx)dx$ and $\int e^{ax}\cos(bx)dx$

52. $\int e^{-x}\sin x\,dx =$
1) $\frac{e^{-x}}{2}[\sin x+\cos x]+c$
2) $\frac{-e^{-x}}{2}[\sin x+\cos x]+c$
3) $-e^{-x}(\sin x+\cos x)+c$
4) $e^{-x}(\sin x+\cos x)+c$

PROBLEMS ON INVERSE TRIGONOMETRIC FUNCTIONS

53. $\int \cot^{-1}\left(\frac{x-1}{x+1}\right)dx =$
1) $\frac{\pi}{4}x + x\cot^{-1}x - \frac{1}{2}\log(1+x^2)+c$
2) $x\cot^{-1}x + \frac{1}{2}\log(1+x^2)+c$
3) $\frac{\pi}{4}x - \frac{1}{2}\log(1+x^2)+c$
4) $\frac{\pi}{4}x + x\cot^{-1}x + \frac{1}{2}\log(1+x^2)+c$

54. $\int \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)dx =$
1) $3\left[x\tan^{-1}x - \frac{1}{2}\log(1+x^2)\right]+c$
2) $x\tan^{-1}x - \frac{1}{2}\log(1+x^2)+c$
3) $3\left[x\tan^{-1}x - \frac{1}{2}\log(1-x^2)\right]+c$
4) $x\tan^{-1}x + c$

55. $\int x\sec^{-1}x\,dx =$
1) $\frac{1}{2}x^2\sec^{-1}x + \frac{1}{2}\sqrt{x^2-1}+c$
2) $\frac{1}{2}x^2\sec^{-1}x - \frac{1}{2}\sqrt{x^2-1}+c$
3) $x^2\sec^2 x + \frac{1}{2}\sqrt{x^2-1}+c$
4) $x^2\sec^2 x - \frac{1}{2}\sqrt{x^2-1}+c$

56. **If** $I_1 = \int \sin^{-1}\left(\frac{2x}{1+x^2}\right)dx$, $I_2 = \int \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)dx$,
$I_3 = \int \tan^{-1}\left(\frac{2x}{1-x^2}\right)dx$ then $I_1 + I_2 - I_3 =$ ____
1) $2x\tan^{-1}x - \log(1+x^2)+c$
2) $2[x\tan^{-1}x - \log(1+x^2)]+c$
3) $2x\tan^{-1}x + \log(1+x^2)+c$ 4) 0

APPLICATIONS OF STANDARD INTEGRALS

57. $\int \frac{\cosh x}{\cosh^2 x+3}dx=$

1) $\mathrm{Tan}^{-1}\left(\frac{\sinh x}{2}\right)+c$ 2) $\frac{1}{2}\mathrm{Tan}^{-1}\left(\frac{\sinh x}{2}\right)+c$

3) $\frac{1}{2}\mathrm{Tan}^{-1}(\sinh x)+c$ 4) $\mathrm{Tan}^{-1}(\sinh x)+c$

58. $\int \frac{1}{x\sqrt{x^6+1}}dx=$

1) $\frac{1}{3}\sinh^{-1}\left(\frac{1}{x^3}\right)+C$ 2)

3) 4)

METHODS OF INTEGRATION

59. $dx=$

1) $\mathrm{Tan}^{-1}(x^3+1)+c$ 2) $\mathrm{Tan}^{-1}(x^3+1)+c$

3) - $\mathrm{Tan}^{-1}(x^3+1)+c$ 4) $\mathrm{Cot}^{-1}(x^3+1)+c$

60.

1)

2)

3) 4)

61. =

1) 2)

3) 4)

62.

1)

2)

3)

4)

63. If

Then

1) 2) 3) 4)

64.

1)

2)

3)

4)

65. If A log |Sin x + Cos x |+Bx + c,

then A =........, B =..........,

1) 2) -1, 1 3) 4)

66. = k Tan^{-1} + c,

then k =

1) 2) 3) 4)

67.

1) 2)

3) 4)

68.

1) 2)

3) 4)

INTEGRATION BY USING PARTIAL FRACTIONS

69.

1) 2)

3) 4)

70.

1) 2)

3) 4)

71.

1) 2)

3) 4)

72. If **then A =**

1) 2 2) 1/2 3) 2/3 4) 1/3

73.

1)

2)

3)

4)

REDUCTION FORMULAE

74. **, then**

1) 2)

3) 4)

75.

1)

2)

3)

4)

76. If for

then

1) 2) 3) 4)

77. If **, and** **then**

1) 2)

3) 4)

78. $\int\left[2\log x+(\log x)^2\right]dx=$

1) $x(\log x)^2+c$ 2) $x^2\log x+c$

3) $x(\log x)^3+c$ 4) $x^3\log x+c$

79. $\int\frac{4\sec^2 x\tan x}{\sec^2 x+\tan^2 x}dx=\log\left|1+f(x)\right|+c$ **then** $f(x)=$

1) $2\sin^2 x$ 2) $2\cos^2 x$

3) $2\tan^2 x$ 4) $2\operatorname{cosec}^2 x$

LEVEL-I (H.W.) - KEY

1)4	2)1	3)4	4)3	5)3	6)1
7)1	8)3	9)1	10)1	11)2	12)3
13)3	14)1	15)4	16)2	17)4	18)3
19) 3	20) 1	21) 4	22) 1	23) 2	24) 4
25)1	26)2	27)1	28)2	29) 3	30)2
31)3	32)2	33)2	34)1	35)1	36)3
37)2	38)1	39)3	40)3	41)2	42)3
43)1	44)3	45)4	46)3	47)2	48)1
49)1	50)1	51)2	52)2	53)4	54)1
55)2	56)1	57)2	58)3	59)2	60)4
61)3	62)1	63)2	64)2	65)1	66)3
67)2	68)3	69)1	70)3	71)3	72) 2
73) 1	74) 3	75) 3	76) 1	77) 2	78) 1
79) 3					

LEVEL-I (H.W.) - HINTS

1. $1+\cos 2x=2\cos^2 x$
2. Divide nr with dr.
3. $\cosh x+\sinh x=e^x$
4. $\sqrt{1-x^4}=\sqrt{1+x^2}\sqrt{1-x^2}$
5. Take n = 1.
6. Req $=\int\cos x dx=\sin x+C$
7. Put $\cos x=t$
8. Put $ax^2+b=t$
9. Put $\cot x=t$
10. Apply the formula $\int\frac{f^1(x)}{f(x)}dx$
11. Apply the formula $\int(f(x))^n f^1(x)dx$
12. Apply the formula $\int\frac{f^1(x)}{f(x)}dx$
13. $\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A\tan B}$
14. Put $\sec x=t$
15. Put $\sin^{-1}(x^2)=t$
16. Put $\sec x+\tan x=t$
17. Put $5x^7=t$
18. Put $\cos(x^2)=t$
19. Put $\sin x=t$
20. Put $\tan x=t$
21. Put $\tan^{-1}\sqrt{x}=t$
22. Put $x^x=t$
23. $4^x=(2^x)^2$ and put $2^x=t$
24. Put $1=\sec^2 x-\tan^2 x$ in the numerator only.
25. I : L.H.S $=\frac{\sin^2 x+\cos^2 x}{\sin x\cos^2 x}$

    II : Divide nr with dr
26. $I=\int\frac{2\cos^2 2x}{\cos^2 x-\sin^2 x}(\sin x\cos x)\,dx$

    $=\frac{1}{2}\int\sin 4x\,dx$.
27. Adding and subtracting 'x' in the numerator
28. Put $1-\frac{1}{x}=t$
29. $\cos x\sqrt{\cos 2x}=\cos^2 x\sqrt{1-\tan^2 x}$
30. Put $x\log x-x=t$.
31. Write $x^3=\left(x^{3/2}\right)^2, a^3=\left(a^{3/2}\right)^2$ and let $x^{3/2}=t$
32. Put $\log(Tan\,x)=t$
33. Verification
34. Use $\int e^x f(x)\,dx$
35. Use $\int e^x f(x)\,dx$

36. $I = \frac{1}{\log_e^{10}} \int \log x \, dx$

37. Put $I = \int \frac{\cos x}{x} dx + \cos x \log x - \int \frac{\cos x}{x} dx$

38. In the interval $(0, \pi)$ $\sin x$ is positive

$\therefore I = \int (3x+4) \sin x \, dx$.

39. $U = \log x, V = x^n$

40. Put $\sin x = t$

41. Put $\log x = t$

42. $\int e^x \left[f(x) + f'(x) \right] dx$

43. $\int e^x \left[f(x) + f'(x) \right] dx$

44. $\int e^x \left[f(x) + f'(x) \right] dx$

45. $\int e^x \left[f(x) + f'(x) \right] dx$

46. Apply the formula $\int e^x \left[f(x) + f^1(x) \right] dx$

47. $t = \log_e x, I = \int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt$

48. Use $\int e^{kx} \left[k\, f(x) + f'(x) \right] dx$

49. Put $x = -t$

50. Put $x = \sin \theta$

51. $\int \left(e^{\tan^{-1} x} + x e^{\tan^{-1} x} \frac{1}{1+x^2} \right) dx$

52. Using the formulae $\int e^{ax} \sin bx dx$ and $\int e^{ax} \cos bx dx$

53. Put $x = \cot \theta$

54. Put $x = \tan \theta$

55. $u = \sec^{-1} x, v = x$

56. Put $x = \tan \theta$

57. $\cosh^2 x = 1 + \sinh^2 x$

58. $\int \frac{1}{x^4 \sqrt{1 + \left(\frac{1}{x^3} \right)^2}} dx$ Put $\frac{1}{x^3} = t$

59. Put $x^3 = t$

60. Put $\mathbf{t = x - \frac{1}{x}}$

61. Put $\sqrt{x} = t$

62. Put $x + 1 = t^2$

63. Put $\tan \frac{x}{2} = t$

64. Put $\tan x = t$

65. Use $\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx = \frac{ac + bd}{c^2 + d^2} x$

$+ \frac{ad - bc}{c^2 + d^2} \log \left| c \cos x + d \sin x \right| + c$

66. $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{ab} \tan^{-1} \left(\frac{a \tan x}{b} \right) + c$

67. Put $\tan x = t$

68. Put $2 \tan x + 3 = t$

69. Put $\cos x = t$

70. Put $\log x = t$

71. Put $x^9 - 1 = t$

72. Resolve into Partial Fractions

73. Resolve into Partial Fractions

74. Apply the formula $I_n = \frac{-\cot^{n-1} x}{n-1} - I_{n-2}$

75. Apply the formula

$I_n = \frac{Sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \cdot I_{n-2}$

76. Put $U = x^n, V = e^{cx}$

77. $\cos(n+1)x - \cos(n-1)x = -2 \sin(nx) \sin x$

78. $\int \left[2 \log x + (\log x)^2 \right] dx$

$\int \left(2t + t^2 \right) e^t = e^t t^2 + c$

$= e^{\log x} (\log x)^2 + c = x (\log x)^2 + c$

79. $\int \frac{4 \sec^2 x \tan x}{1 + 2 \tan^2 x} dx$ put $2 \tan^2 x = t$